United States Patent Application Publication	20250280225
Kind Code	A1
Publication Date	September 04, 2025
Inventor(s)	XIE; Shuailin et al.

WEARING ASSEMBLIES AND HEADPHONES

Abstract

A wearing assembly and an earphone are provided. The wearing assembly includes an elastic metal wire, a lead assembly, an insert member, and a wrapping assembly. The insert member is fixed to an end of the elastic metal wire and includes an insert body. The wrapping assembly includes a body wrapping portion wrapping a periphery of the elastic metal wire and the lead assembly, the lead assembly has an exposed portion that is exposed from the body wrapping portion and extends to the insert body; and at least one of the insert body and the wrap assembly further wraps a periphery of the exposed portion. The above solution uses at least one of the insert body or the wrap assembly to wrap the periphery of the exposed portion to improve a mounting yield of the wearing assembly.

Inventors: **XIE; Shuailin (Shenzhen, CN), GUO; Burui (Shenzhen, CN), LI; Chaowu (Shenzhen, CN)**

Applicant: **SHENZHEN SHOKZ CO., LTD.** (Shenzhen, CN)

Family ID: **91784493**

Assignee: **SHENZHEN SHOKZ CO., LTD. (Shenzhen, CN)**

Appl. No.: **19/211270**

Filed: **May 18, 2025**

Foreign Application Priority Data

CN	202310541798.X	May. 12, 2023

Related U.S. Application Data

parent WO continuation PCT/CN2023/111837 20230808 PENDING child US 19211270

Publication Classification

Int. Cl.: **H04R1/10** (20060101)

U.S. Cl.:

CPC **H04R1/1066** (20130101); **H04R1/1008** (20130101); **H04R1/105** (20130101); H04R2460/13 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of International Application No. PCT/CN2023/111837 filed on Aug. 8, 2023, which claims priority of Chinese Patent Application No. 202310541798.X, filed on May 12, 2023, the contents of each of which are entirely incorporated herein by reference.

TECHNICAL FIELD
[0002] The present disclosure relates to the technical field of electronic devices, and in particular to a wearing assembly and a headphone.
BACKGROUND
[0003] With the continuous popularization of an electronic device, the electronic device is becoming an indispensable social and entertainment tool in people's daily life. The electronic device such as a headphone is widely used in people's daily life, which is used in conjunction with a cell phone, a computer, and other terminal devices to facilitate a provision of an auditory feast for a user.
[0004] Taking the headphone as an example, the headphone may be disposed with a wearing assembly for the user to wear, and the wearing assembly often needs to include a lead. When the wearing assembly is fixed with other portions, if an adhesion between the lead and an adhesive used is poor, a fixing effect may be poor too.
SUMMARY
[0005] The present disclosure provides a wearing assembly and a headphone. The wearing assembly includes an elastic metal wire, a lead assembly, an insert member, and a wrapping assembly. The insert member is fixed to an end of the elastic metal wire and includes an insert body, the wrapping assembly includes a body wrapping portion wrapping a periphery of the elastic metal wire and the lead assembly, the lead assembly has an exposed portion that is exposed from the body wrapping portion and extends to the insert body; at least one of the insert body and the wrap assembly further wraps a periphery of the exposed portion.
[0006] In some embodiments, the insert body is provided with a hole path, and the exposed portion penetrates the hole path.
[0007] In some embodiments, a hole wall of the hole path surrounds a periphery of the exposed portion along a circumferential direction of the exposed portion in a closed manner.
[0008] In some embodiments, the wearing assembly further includes an epitaxial wrapping portion integrally molded with the body wrapping portion, the epitaxial wrapping portion wraps on an end of the insert body facing the body wrapping portion along a circumferential direction of the insert body and the other end of the insert body away from the body wrapping portion is exposed.
[0009] In some embodiments, a wrapping length of the epitaxial wrapping portion over the insert body in an insertion direction relative to the insert body is in a range of 0.2 mm-2.0 mm.
[0010] In some embodiments, the insert member further includes an epitaxial positioning portion protruding from an end surface of the insert body towards the body wrapping portion, and the

epitaxial positioning portion is embedded within the body wrapping portion.
[0011] In some embodiments, the elastic metal wire is threaded within the epitaxial positioning portion.
[0012] In some embodiments, in a cross section perpendicular to an insertion direction of the insert body relative to a first insert groove, the insert body has a lengthwise direction and a widthwise direction, and the exposed portion is spaced apart from the elastic metal wire along the lengthwise direction or the widthwise direction.
[0013] In some embodiments, the insert body is provided with two slots spaced apart along the lengthwise direction, and the exposed portion and the elastic metal wire are disposed between the two slots in the lengthwise direction.
[0014] In some embodiments, the wrapping assembly further includes an epitaxial wrapping portion wrapping the periphery of the exposed portion of the lead assembly and partially covering the insert body in a circumferential direction of the insert body.
[0015] In some embodiments, a thickness of the epitaxial wrapping portion is less than a thickness of the body wrapping portion, and the epitaxial wrapping portion is connected to the body wrapping portion to form a step structure.
[0016] In some embodiments, the epitaxial wrapping portion is integrally molded with the body wrapping portion.
[0017] In some embodiments, a receiving groove is disposed on a surface of the insert body, and in a cross section perpendicular to an insertion direction of the insert body relative to a first insert groove, the receiving groove has an opening, the exposed portion is disposed within the receiving groove, the epitaxial wrapping portion is integrally molded with the body wrapping portion, and at least a part of the exposed portion disposed within the receiving groove has an externally visible region when viewed from the opening of the receiving groove, and the externally visible region is wrapped by the epitaxial wrapping portion.
[0018] In some implementations, a lengthwise direction of the receiving groove is provided along the insertion direction.
[0019] In some embodiments, the epitaxial wrapping portion further fills in a gap between the exposed portion and a groove wall of the receiving groove.
[0020] In some implementations, the wrapping assembly further includes an epitaxial insert portion integrally molded with the body wrapping portion, and the epitaxial insert portion contacts an end surface of the insert body towards the body wrapping portion.
[0021] In some embodiments, the insert member further includes an epitaxial positioning portion protruding from the end surface of the insert body, and the epitaxial positioning portion is embedded within the epitaxial insert portion.
[0022] In some embodiments, the insert member further includes an epitaxial positioning portion protruding from an end surface of the insert body towards the body wrapping portion, and the epitaxial positioning portion is embedded within the body wrapping portion.
[0023] In some embodiments, the elastic metal wire is threaded within the epitaxial positioning portion.
[0024] In some embodiments, in a cross section perpendicular to an insertion direction of the insert body relative to a first insert groove, the insert body has a lengthwise direction and a widthwise direction, and the exposed portion is spaced apart from the elastic metal wire along the lengthwise direction or the widthwise direction.
[0025] In some embodiments, the insert body is disposed with two slots spaced apart in the lengthwise direction, and the exposed portion and the elastic metal wire are disposed between the two slots in the lengthwise direction.
[0026] In some embodiments, a covering width of the epitaxial wrapping portion over the insert body in the circumferential direction of the insert body is less than or equal to one-fourth of a maximum circumferential length of the insert body.

[0027] The present disclosure also provides a headphone including the wearing assembly as described above, and a housing assembly disposed with a first insert groove, and the insert body is inserted in the first insert groove.
[0028] In some embodiments, the headphone further includes a sealant configured to seal a gap between at least one of the insert body or the wrapping assembly which wraps the exposed portion and a groove wall of the first insert groove. Surface wettability of the sealant on a surface of the at least one of the insert body and the wrapping assembly which wraps the exposed portion is greater than surface wettability of the sealant on a surface of the lead assembly.
[0029] In some embodiments, a material of the insulating coating is polytetrafluoroethylene (PTFE), and a material of the insert body is metal.
[0030] Beneficial effects of the present disclosure is that: by increasing at least one of the wrapping assembly and the insert body to protect the exposed portion, the adhesive cannot directly contact the exposed portion of the lead assembly, and by directly bonding at least one of the wrapping assembly and the insert body to the adhesive, a bonding effect of adhesive can be enhanced, thereby improving a fixing effect and a sealing effect, and a mounting yield of the wearing assembly can be improved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0031] The more clearly illustrate the technical solutions in the embodiments of the present disclosure, the following will briefly introduce the accompanying drawings that need to be used in the description of the embodiments, and it is obvious that the accompanying drawings in the following description are only some of the embodiments of the present disclosure, and other drawings may also be obtained according to these drawings without creative labor for those skilled in the art.
[0032] FIG. **1** is a schematic diagram illustrating a general assembly three-dimensional structure of a headphone according to an embodiment of the present disclosure;
[0033] FIG. **2** is an exploded structural diagram illustrating a movement assembly shown in FIG. **1**;
[0034] FIG. **3** is a schematic diagram illustrating a cross section of an earhook assembly shown in FIG. **1**;
[0035] FIG. **4** is a schematic diagram illustrating a decomposition structure of the earhook assembly shown in FIG. **3**;
[0036] FIG. **5** is a schematic diagram illustrating another cross section of the earhook assembly shown in FIG. **3**;
[0037] FIG. **6** is a schematic diagram illustrating an enlarged structure of a J region of the earhook assembly shown in FIG. **5**;
[0038] FIG. **7** is a schematic diagram illustrating a cross section of the earhook assembly shown in FIG. **3** with some parts hidden;
[0039] FIG. **8** is a schematic diagram illustrating an enlarged structure of a K region of the earhook assembly shown in FIG. **7**;
[0040] FIG. **9** is a schematic diagram illustrating another decomposition structure of the earhook assembly shown in FIG. **1**;
[0041] FIG. **10** is a schematic diagram illustrating a cross section of the earhook assembly shown in FIG. **9**;
[0042] FIG. **11** is a schematic diagram illustrating an enlarged structure of an R region of the earhook assembly shown in FIG. **10**;
[0043] FIG. **12** is a schematic diagram illustrating another cross section of the earhook assembly shown in FIG. **9** with some parts hidden;

[0044] FIG. **13** is a schematic diagram illustrating an enlarged structure of a T region of the earhook assembly shown in FIG. **12**;
[0045] FIG. **14** is a schematic diagram illustrating a partial structure of the headphone shown in FIG. **1**;
[0046] FIG. **15** is a schematic diagram illustrating a decomposition structure of the structure shown in FIG. **14**;
[0047] FIG. **16** is a schematic diagram illustrating a cross section of the structure shown in FIG. **14** along a P-P section line;
[0048] FIG. **17** is a schematic diagram illustrating a cross section of the structure shown in FIG. **14** along an R-R section line;
[0049] FIG. **18** is a schematic diagram illustrating another cross section of the structure shown in FIG. **14** along a P-P section line;
[0050] FIG. **19** is a schematic diagram illustrating an enlarged structure of a Q region shown in FIG. **16**;
[0051] FIG. **20** is a schematic diagram of a partial structure of a wearable component shown in FIG. **15**;
[0052] FIG. **21** is a schematic diagram illustrating partial structures of a stopper and an insert body shown in FIG. **15**;
[0053] FIG. **22** is a schematic diagram illustrating another partial structure of the headphone shown in FIG. **1**;
[0054] FIG. **23** is a schematic diagram illustrating a decomposition structure shown in FIG. **22**;
[0055] FIG. **24** is a schematic diagram illustrating a cross section of the structure shown in FIG. **22** along an E-E section line;
[0056] FIG. **25** is a schematic diagram illustrating an enlarged structure of a Q region shown in FIG. **24**;
[0057] FIG. **26** is a schematic diagram illustrating a cross section of the structure shown in FIG. **22** along a G-G section line;
[0058] FIG. **27** is a schematic diagram illustrating an enlarged structure of an H region of the headphone shown in FIG. **26**;
[0059] FIG. **28** is a schematic diagram illustrating a partial structure of the headphone shown in FIG. **1**;
[0060] FIG. **29** is a schematic diagram illustrating a cross section of the structure shown in FIG. **28** along an M-M section line;
[0061] FIG. **30** is a schematic diagram illustrating a structure of a housing assembly shown in FIG. **29**;
[0062] FIG. **31** is a schematic diagram illustrating an enlarged structure of an A**1** region shown in FIG. **29**;
[0063] FIG. **32** is a schematic diagram illustrating a structure of a wearable assembly according to an embodiment of the present disclosure;
[0064] FIG. **33** is a schematic diagram illustrating an enlarged structure of an A**2** region in FIG. **29**;
[0065] FIG. **34** is a schematic diagram illustrating an exemplary partial structure of a rear hanging assembly shown in FIG. **1**;
[0066] FIG. **35** is a schematic diagram illustrating another view of a structure of the rear hanging assembly shown in FIG. **34**;
[0067] FIG. **36** is a schematic diagram illustrating a cross section of the rear hanging assembly shown in FIG. **34** along an insertion direction;
[0068] FIG. **37** is a schematic diagram illustrating another exemplary partial structural of the rear hanging assembly shown in FIG. **1**;
[0069] FIG. **38** is a schematic diagram illustrating a cross section of a lead assembly of the rear hanging assembly shown in FIG. **34**;

[0070] FIG. **39** is a schematic diagram illustrating another exemplary partial structure of the rear hanging assembly shown in FIG. **1**; and

[0071] FIG. **40** is a schematic diagram illustrating a structure of an insert member of the rear hanging assembly shown in FIG. **39**.

DETAILED DESCRIPTION

[0072] The present disclosure is described in further detail below in conjunction with the accompanying drawings and embodiments. In particular, it is noted that the following embodiments are only used to illustrate the present disclosure, but do not limit the scope of the present disclosure. Similarly, the following embodiments are only part of the embodiments of the present disclosure rather than all of the embodiments, and all other embodiments obtained by those skilled in the art without creative labor fall within the scope of protection of the present disclosure.

[0073] References to “embodiments” in the present disclosure mean that particular features, structures, or characteristics described in conjunction with the embodiments are included in at least one embodiment of the present disclosure. It is understood by those skilled in the art both explicitly and implicitly that the embodiments described in the present disclosure are combined with other embodiments.

[0074] The following embodiment of the present disclosure describes an exemplary structure of a headphone **100**.

[0075] As shown in FIG. **1**, the headphone **100** may include a movement assembly **1**, an earhook assembly **2**, and a rear hanging assembly **3**. There are two movement assemblies **1**. The two movement assemblies **1** are configured to transmit a vibration and/or a sound to left and right ears of a user, respectively. The two movement assemblies **1** may be the same or different. For example, one movement assembly **1** is disposed with a microphone, and the other movement assembly **1** is not disposed with the microphone. As another example, one movement assembly **1** is disposed with a key and a corresponding circuit board, and the other movement assembly **1** is not disposed with the key and the corresponding circuit board. The two movement assemblies **1** may be the same on a movement module (e.g. a speaker module). The movement assembly **1** described subsequently herein may be regarded as taking one of the two movement assemblies **1** as an example for a detailed illustration. There may be two earhook assemblies **2**, and the two earhook assemblies **2** are located in the left and right ears of the user, respectively, so as to enable the movement assembly **1** to fit a face of the user. For example, one earhook assembly **2** is disposed with a battery, and the other earhook assembly **2** is disposed with a control circuit, etc. One end of the earhook assembly **2** is connected to the movement assembly **1**, and the other end of the earhook assembly **2** is connected to the rear hanging assembly **3**. The rear hanging assembly **3** connects the two ear hook assemblies **2**, and the rear hanging assembly **3** is configured to wrap around a back of the neck or the head of the user and provides a clamping force so that the two movement assemblies **1** are clamped to both sides of a face of the user and the earhook assemblies **2** are more fixedly hang on the ears of the user. Of course, the headphone **100** may not include the rear hanging assembly **3**, and the movement assemblies **1** are worn in the ears of the user via the earhook assemblies **2**.

[0076] The following mainly describes an exemplary structure of the movement assembly **1** of the headphone **100**, etc.

[0077] As shown in FIG. **2**, the movement assembly **1** may include a housing assembly **10**, a bone conduction speaker **11**, and an air conduction speaker **12**.

[0078] In some embodiments, the housing assembly **10** may be disposed with a receiving cavity **1001** and a receiving cavity **1002** that are isolated from each other. The housing assembly **10** may also be referred to as the movement housing assembly **10**. The bone conduction speaker **11** is disposed within the receiving cavity **1001**.

[0079] The air conduction speaker **12** conducts a sound into an ear canal of the user by means of air vibration, and the bone conduction speaker **11** conducts the sound to the user by means of bone conduction vibration. In some embodiments, a sealing of the receiving cavity **1001** is greater than a sealing of the receiving cavity **1002**. The sealing may be regarded as an air tightness. As the receiving cavity **1002** in which the air conduction speaker **12** is placed needs to be connected to an outside world to facilitate a conduction of sound waves through air, the bone conduction speaker **11** and the air conduction speaker **12** are disposed independently from each other, and there is no need to dispose the bone conduction speaker **11** in the receiving cavity **1002**. As the receiving cavity **1001** is set independently, the sealing of the accommodating cavities **1001** may be at a higher level, which effectively improves a sealing effect of the bone conduction speaker **11** and thereby preventing the bone conduction speaker **11** from being damaged by erosion of external environmental factors, and at the same time ensures a sound quality of the air conduction speaker **12**. In addition, the headphone **100** may effectively reduce a mutual interference between the bone conduction speaker **11** and the air conduction speaker **12** when the bone conduction speaker **11** and the air conduction speaker **12** are working at the same time, thereby effectively improving the sound quality of the headphone **100**.

[0080] In some embodiments, the housing assembly **10** includes a housing **101**, a housing **102**, and a housing **103**, with the housing **101** and the housing **102** cooperating with each other to form the receiving cavity **1001**. The housing **101** and/or the housing **102** further form a portion of the receiving cavity **1002**, and the housing **103** may form another portion of the receiving cavity **1002**, and further, the housing **103** and the housing **101** and/or the housing **102** cooperate with each other to form the receiving cavity **1002**. In some embodiments, at least the housing **101** and the housing **102** cooperate together to form the receiving cavity **1001**, and at least the housing **103** and the housing **101** cooperate together to form the receiving cavity **1002**.

[0081] In some embodiments, a vibration direction of the bone conduction speaker **11** is disposed in a crosswise direction with a vibration direction of the air conduction speaker **12**, and the housing **101** and the housing **102** cooperate with each other along the vibration direction of the bone conduction speaker **11**. The housing **103** cooperates with the housing **101** and/or the housing **102** along the vibration direction of the air conduction speaker **12**.

[0082] The vibration direction of the bone conduction speaker **11** may also be referred to herein as a bone conduction vibration direction X**1**, and the vibration direction of the air conduction speaker **12** may also be referred to herein as an air conduction vibration direction X**2**. The bone conduction vibration direction X**1** and the air conduction vibration direction X**2** are set at a cross instead of being parallel to each other. For example, the bone conduction vibration direction X**1** and the air conduction vibration direction X**2** are set perpendicularly or approximately perpendicular to each other (e.g.,) 90°±10°. When the bone conduction speaker **11** and the air conduction speaker **12** are working simultaneously, the bone conduction speaker **11** and the air conduction speaker **12** are vibrating along the bone conduction vibration direction X**1** and the air conduction vibration direction X**2**, respectively. As the two vibration directions are set crosswise, an impact of the vibration of the bone conduction speaker **11** on the sound quality of the air conduction speaker **12** due to the vibrations of the bone conduction speaker **11** and the air conduction speaker **12** in the same direction may be effectively relieved.

[0083] Referring to FIG. **2**, in some embodiments, the housing assembly **10** may also be disposed with a pressure relief hole **1081** for connecting the receiving cavity **1002** and the external environment. The pressure relief hole **1081** is disposed to extend toward a side where the receiving cavity **1001** is located. The housing assembly **10** is disposed with the pressure relief hole **1081** for relieving pressure for the air conduction speaker **12**. The pressure relief hole **1081** connects the receiving cavity **1002** to the external environment, and the pressure relief hole **1081** extends from a connection with the receiving cavity **1002** toward a side where the receiving cavity **1001** is located.

[0084] The housing assembly **10** may include an interval wall **1012** for separating the receiving cavity **1001** from the receiving cavity **1002**, with at least a portion of the interval wall **1012** extending toward the side where the receiving cavity **1001** is located and forming a portion of a hole wall of the pressure relief hole **108**. The interval wall **1012** is disposed on the housing **101**, and the housing **101** is disposed with a sub-receiving cavity **1010** and a sub-receiving cavity **1011** disposed on opposite sides of the interval wall **1012**, and an opening direction of the sub-receiving cavity **1010** is along a wall surface of the interval wall **1012**. Specifically, the opening direction of the sub-receiving cavity **1010** is parallel or substantially parallel to the wall surface of the interval wall **1012**. The opening direction of the sub-receiving cavity **1011** is disposed crosswise with the wall surface of the interval wall **1012**. The housing **102** is provided with a sub-receiving cavity **1020**, the housing **102** caps an open end of the sub-receiving cavity **1010**, and the sub-receiving cavity **1020** cooperates with the sub-receiving cavity **1010** to form the receiving cavity **1001**. The housing **103** is provided with a sub-cavity **1030**, the housing **103** caps an open end of the sub-cavity **1011**, and the sub-cavity **1030** cooperates with the sub-cavity **1011** to form the receiving cavity **1002**.

[0085] The following proceeds to describe primarily the structure of the earhook assembly **2**, etc., of the headphone **100**.

[0086] Combining FIGS. **3-5**, the earhook assembly **2** may include a housing assembly **20** and a key assembly **21**. The housing assembly **20** includes a housing **201** and a housing **202**. The housing **201** is provided with a receiving cavity **2010** with at least an open end **2011**. The key assembly **21** is at least partially inserted into the receiving cavity **2010** through the open end **2011**, and the key assembly **21** cooperates with the housing **201** to form an adhesive receiving groove **210** that is visible externally from the housing **201** through the open end **2011**. The housing **202** is disposed over the open end **2011** of the housing **201** and covers the adhesive receiving groove **210**. The open end **2011** refers to an end of the housing **201** disposed with an opening.

[0087] In some embodiments, a circuit board **28** is disposed within the receiving cavity **2010**. The circuit board **28** is configured to perform a conversion processing on an electrical signal to support the implementation of various functions of the headphone **100**. The key assembly **21** is configured to accept a press of the user to be active relative to the circuit board **28**, thereby triggering the circuit board **28** to operate. For example, the key assembly **21** triggers the circuit board **28** to realize the functions such as switching on/off, switching playback content, increasing/decreasing a sound volume, etc., of the headphone **100** when pressed by the user.

[0088] Combining FIG. **9**, a battery **29** may also be disposed within the receiving cavity **2010**. In some embodiments, the battery **29** is disposed in one of the two housing assemblies **20** of the earhook assembly **2**, and the circuit board **28** is disposed in another one of the two housing assemblies **20**. The two housing assemblies **20** may be the same or different.

[0089] In some embodiments, one side of the circuit board **28** may be connected to a lead within the receiving cavity **2010**, and the other side of the circuit board **28** is near the open end **2011** and abuts against the key assembly **21**. The circuit board **28** may be assembled into the receiving cavity **2010** through the open end **2011**. The circuit board **28** may be limited within the receiving cavity **2010** by capping the housing **202** over the open end **2011** of the housing **201**. By disposing the housing **202** to cover the adhesive receiving groove **210**, the headphone **100** may be made more aesthetically pleasing. The housing **202** may also limit a portion of the key assembly **21** within the receiving cavity, thereby limiting the key assembly **21** from detaching from the housing **201**. In some embodiments, the housing **202** partially or fully covers the adhesive receiving groove **210**.

[0090] A adhesive may be disposed between the key assembly **21** and the housing **201**. Here, the adhesive may be a generic term, and the adhesive may be of various forms, such as a solid adhesive, a liquid adhesive, a semi-solid adhesive, etc., which is configured to serve as a sealing or fixing function. The adhesive may form a sealing between the key assembly **21** and the housing **201** to improve a waterproof performance of the headphone **100**. The adhesive may also act as an adhesive fixing. When dispensing the adhesive, the adhesive, for example, may be in the form of a fluid, and the adhesive may be cured after a period of time.

[0091] If the adhesive receiving groove **210** is not disposed, the adhesive needs to be pre-applied between the key assembly **21** and the housing **201** along the insertion direction before an insert mounting of the key assembly **21** is performed. When mounting the key assembly **21**, the key assembly **21** is likely to squeeze the adhesive into the receiving cavity **2010**, thereby affecting a normal operation of the headphone **100**. The adhesive receiving groove **210** may be configured to hold the adhesive. By disposing the adhesive receiving groove **210**, if the dispensing of adhesive is performed after the insertion and mounting of the key assembly **21**, the adhesive is effectively prevented from being pressed into the receiving cavity **2010** by the key assembly **21**. By disposing the adhesive receiving groove **210** to be visible from the outside of the housing **201** through the open end **2011**, it is easy to perform the dispensing operation of the adhesive, and to control an amount of adhesive dispensed to reduce a risk of the headphone **100** being poorly sealed due to the amount of adhesive dispensed being too small, and to reduce a probability of the adhesive seeping out to an outer surface of the housing **201** or seeping into the receiving cavity **2010** due to too much adhesive being dispensed, so as to minimize an adverse effect on the appearance of the headphone **100** and an operation state of the headphone **100**.

[0092] As shown in FIG. **4**, FIG. **6**, and FIG. **8**, the key assembly **21** may include a key support **211**, an annular protrusion **2110** is disposed on an outer wall surface of the key support **211**, an annular support platform **2012** is disposed on an inner wall surface of the housing **201**, the annular protrusion **2110** is supported on the annular support platform **2012**, and the annular protrusion **2110** sinks for a certain distance relative to the open end **2011**. On a side of the annular protrusion **2110** towards the open end **2011**, the outer wall surface of the key support **211** and the inner wall surface of the housing **201** cooperate with each other to form the adhesive receiving groove **210**.

[0093] By disposing the annular protrusion **2110** to be supported on the annular support platform **2012**, the annular support platform **2012** limits a movement of the key support **211** toward the receiving cavity **2010**, enabling the key support **211** to be stably supported on the housing **201**. When assembling the key support **211**, the annular support platform **2012** may play a role in positioning the key support **211**, making it easy for the key support **211** to be mounted in place, thereby improving an assembly efficiency. Additionally, the annular protrusion **2110** is supported on the annular support platform **2012** to make a flow path for the adhesive flowing into the receiving cavity **2010** to be narrow and zigzagging, which reduces the flow of the adhesive from the adhesive receiving groove **210** to the receiving cavity **2010** when dispensing the adhesive.

[0094] By disposing the annular protrusion **2110** to sink for a certain distance relative to the open end **2011**, the annular protrusion **2110** may form a bottom wall of the adhesive receiving groove **210**. In contrast to forming the adhesive receiving groove **210** in an integrally molded manner, forming the adhesive receiving groove **210** in a split molding manner through the key support **211** and the housing **201** being abut against each other makes the assembly of the headphone **100** more flexible and thus easier to assemble.

[0095] In some embodiments, as shown in FIG. **5** and FIG. **6**, an outer end surface of the key support **211** on a side away from the receiving cavity **2010** protrudes from the open end **2011**, and the key assembly **21** includes an elastic pressing member **212** disposed on the outer end surface, a housing **202** is disposed with a keyhole **2020**, and the elastic pressing member **212** is exposed through the keyhole **2020**. By disposing the outer end surface of the side of the key support **211** away from the receiving cavity **2010** protruding from the open end **2011**, the flow of adhesive towards the key assembly **21** may be limited when the adhesive overflows from the adhesive receiving groove **210**.

[0096] When the user presses the side of the key assembly **21** away from the receiving cavity **2010**, the elastic pressing member **212** may elastically deform. The elastic pressing member **212** may be restored when the user releases the key assembly **21**. The pressing action of the user may be transmitted to a switching circuitry on the circuit board **28** through the elastic deformation or deformation restoration of the elastic pressing member **212**, thereby triggering the circuit board **28** to operate. The elastic deformation or deformation restoration of the elastic pressing member **212** may be allowed by disposing the keyhole **2020**, which also allows the press action exerted by the user to be transmitted to the circuit board **28**.

[0097] In some embodiments, as shown in FIG. **5**, the key assembly **21** includes an outer pressing member **213**. A portion of the outer pressing member **213** is inserted into the keyhole **2020**, and another portion of the outer pressing member **213** protrudes from the outer surface of the housing **202** to be pressed by the user. The outer pressing member **213** abuts against the elastic pressing member **212**, and the pressing action of the user may be transmitted to the elastic pressing member **212** through the outer pressing member **213**.

[0098] As shown in FIG. **5** and FIG. **6**, in the insertion direction of the key assembly **21** relative to the receiving cavity **2010**, the housing **201** includes a first wall portion **2013** and a second wall portion **2014** connected to each other, with the second wall portion **2014** located on a side of the first wall portion **2013** away from the open end **2011**, and the annular support platform **2012** is located at a connection between the first wall portion **2013** and the second wall portion **2014**. Specifically, the inner wall surface of the first wall portion **2013** and the inner wall surface of the second wall portion **2014** are connected by the annular support platform **2012**. The key support **211** includes an insert portion **2111** disposed on the side of the annular protrusion **2110** away from the open end **2011**, and the insert portion **2111** is inserted into a space enclosed by the second wall portion **2014**.

[0099] In some embodiments, an outer peripheral dimension of the insertion portion **2111** may match an inner peripheral dimension of the second wall portion **2014**. The second wall portion **2014** may act to position the insert portion **2111**, limit a relative movement between the key support **211** and the housing **201**, so that the adhesive receiving groove **210** stably exists, which is conducive to dispensing the adhesive.

[0100] As shown in FIG. **5** and FIG. **6**, the housing assembly **20** further includes an elastic wrapping layer **203** wrapped on the outer surface of the housing **201**. On a side of the open end **2011**, the housing **201** has an exposed portion **201*a*** that is not wrapped by the elastic wrapping layer **203**. The exposed portion **201*a*** is inserted within the housing **202**, and the housing **202** is abut against the elastic wrapping layer **203** in an insertion direction of the exposed portion **201*a*** relative to the housing **202**.

[0101] By disposing the elastic wrapping layer **203** on the outer surface of the housing **201**, a comfort of the user when holding the headphone **100** is improved, and a friction is increased. By inserting the exposed portion **201*a*** of the housing **201** into the housing **202**, the housing **201** may act to position the housing **202**, which limits the relative movement between the housing **201** and the housing **202** and improves a connection stability between the housing **201** and the housing **202**.

[0102] By disposing the exposed portion **201*a*** of the housing **201** as not being wrapped by the elastic wrapping layer **203**, the elastic wrapping layer **203** may be avoided from interfering with the connection between the housing **201** and the housing **202**. By disposing the housing **202** to abut against the elastic wrapping layer **203** in the insertion direction of the exposed portion **201*a*** relative to the housing **202**, the outer surface of the housing **202** and the outer surface of the elastic wrapping layer **203** are made to flush at the abutment portion, which on the one hand makes the housing **202** and the elastic wrapping layer **203** more tightly packed at the abutment portion, which plays a better sealing effect, thereby reducing an entry of debris and water vapor, and improves a service life of the elastic wrapping layer **203**, and on the other hand the flushness at the abutment portion reduces a discomfort caused by the user when touched, which makes the headphone **100** more integrally stronger and more comfortable, and also makes the headphone **100** more aesthetically pleasing in appearance.

[0103] In some embodiments, the housing **202** is not wrapped by the elastic wrapping layer **203**, so that the side of the key assembly **21** away from the receiving cavity **2010** is spaced apart from the elastic wrapping layer **203**, thereby avoiding the elastic wrapping layer **203** from interfering with the pressing of the user on the key assembly **21** which affects a pressing feeling of the user.

[0104] As shown in FIG. **5** and FIG. **6**, the exposed portion **201*a*** cooperates with the housing **202**

to form an adhesive receiving gap **2003** for connecting the adhesive receiving groove **210**. The adhesive receiving gap **2003** may receive adhesive so that a seal is formed between the exposed portion **201***a* and the housing **202**, and the exposed portion **201***a* is fixed to the housing **202**.
[0105] In some embodiments, the outer end surface of the key support **211** on a side away from the receiving cavity **2010** is set to protrude from the open end **2011**, allowing the adhesive to flow to the adhesive receiving gap **2003** as the adhesive overflows the adhesive receiving groove **210**. In some embodiments, before assembling the housing **201** with the housing **202**, the dispensing of adhesive may be performed on an outer side of the exposed portion **201***a* or on an inner side of the housing **202**, such that the adhesive fills the adhesive receiving gap **2003** after the exposed portion **201***a* and the housing **202** cooperate to form the adhesive receiving gap **2003**.
[0106] As shown in FIG. **5** and FIG. **6**, a depth H**1** of at least a portion of the adhesive receiving groove **210** is greater than or equal to 0.2 mm in the insertion direction of the key assembly **21** relative to the adhesive receiving cavity **2010**. For example, the depth H**1** is 0.25 mm, 0.3 mm, 0.35 mm, 0.4 mm, or 0.5 mm. In some embodiments, a width L**1** of at least a portion of the adhesive receiving groove **210** is greater than or equal to 0.4 mm in a vertical direction of the insertion direction of the key assembly **21** relative to the receiving cavity **2010**. For example, the width L**1** is 0.45 mm, 0.5 mm, 0.55 mm, 0.6 mm or 0.8 mm.
[0107] In this way, on the one hand, there is sufficient adhesive in the adhesive receiving groove **210**, making the adhesive have a sufficient sealing or fixing effect, on the other hand, on a basis that a receiving volume of the adhesive receiving groove **210** is able to hold sufficient adhesive, as the adhesive receiving gap **2003** is connected to the adhesive receiving groove **210**, a small amount of adhesive is allowed to overflow into the adhesive receiving gap **2003**, which further seals between the housing **201** and the housing **202**, thereby enhancing a sealing performance, and as the adhesive receiving gap **2003** is able to retain the adhesive, it relives a risk of the adhesive further oozing out to the outer surface of the housing **201** in an excessive manner.
[0108] As shown in FIG. **3** and FIG. **7**, the housing **201** is further disposed with an accommodation groove **2004** that is connected to the receiving cavity **2010**. The headphone **100** further includes an interface assembly **22** inserted into the accommodation groove **2004**, and the housing **202** is disposed to partially cover the interface assembly **22**.
[0109] The interface assembly **22** may be configured to connect to an external power source to charge the headphone **100**. The interface assembly **22** may also be configured to connect an external device to transfer data between the external device and the headphone **100**. For example, the interface assembly **22** is a Type-C interface or other types of interfaces, such as a mini-USB interface. In some embodiments, the interface assembly **22** is connected to the circuit board **28** within the receiving cavity **2010**. The external device or the external power sources are connected to the interface assembly **22** through the accommodation groove **2004**.
[0110] By disposing the housing **202** to partially shield the interface assembly **22**, the housing **202** is able to stop the interface assembly **22**, which limits the movement of the interface assembly **22** outwardly towards the housing **202** in the direction away from the receiving cavity **2010**, which is conducive to a structural stability of the headphone **100**, and at the same time facilitates the assembly of the interface assembly **22**.
[0111] As shown in FIG. **3**, FIG. **7**, and FIG. **8**, the headphone **100** further includes a dust plug **23**, the dust plug **23** includes a plug body **231** and a fixing portion **232**. The fixing portion **232** is disposed on the housing **201** between the key assembly **21** and the interface assembly **22** through the housing **202**, and the plug body **231** is connected to the fixing portion **232** and is configured to selectively shield the interface assembly **22**.
[0112] The dust plug **23** protects the interface assembly **22** when it shields the interface assembly **22**, for example, the dust plug **23** may keep water and dust away from the interface assembly **22**, and also reduce a risk of mechanical damage to the interface assembly **22**.
[0113] In some embodiments, the dust plug **23** has a deformation capability, and according to the

deformation of the dust plug **23**, the plug body **231** and the fixing portion **232** move relative to each other. By moving relative to the fixing portion **232**, the plug body **231** may be switchable between shielding the interface assembly **22** and not shielding the interface assembly **22**. Specifically, the plug body **231** may form a shield against the interface assembly **22** by being inserted into the accommodation groove **2004**.

[0114] As shown in FIG. **7**, the fixing portion **232** may extend through the housing **202** and onto the housing **201** between the key assembly **21** and the interface assembly **22**, thereby reducing a space taken up by the fixing portion **232** on an outer surface of the housing **202** to improve a space utilization. Specifically, the fixing portion **232** may have a recess **2321**, and the housing **202** may be provided with a through hole **202***a*, through which the fixing portion **232** is threaded through the housing **202**. A hole wall of the through hole **202***a* may be snapped into the recess **2321**, and thus limited within the recess **2321**, such that a bottom wall of the recess **2321** is disposed relative to the hole wall of the through hole **202***a*, which limits a relative movement between the housing **202** and the fixing portion **232**.

[0115] The dust plug **23** may be prepared of an elastic material, such as silicone, rubber, etc., so as to facilitate the insertion of the plug body **231** into the accommodation groove **2004**, and to facilitate the threading of the fixing portion **232** through the housing **202**, and at the same time to make the dust plug **23** deformable.

[0116] As shown in FIGS. **9-11**, the earhook assembly **2** may include a housing assembly **26**. The housing assembly **26** includes a housing **261** and a housing **262** that cooperate with each other, with an adhesive receiving groove **2601** disposed in a first of the housing **261** and the housing **262**. The adhesive receiving groove **2601** includes at least a first sidewall **2602** near an interior of the housing assembly **26**, a second sidewall **2603** away from the interior of the housing assembly **26**, and a bottom wall connecting the first sidewall **2602** and the second sidewall **2603**. A second of the housing **261** and the housing **262** is inserted into the adhesive receiving groove **2601** to form a first gap **2604** with the first sidewall **2602** and form a second gap **2603** with the second sidewall **2603**. A width L**2** of the first gap **2604** is greater than a width L**3** of the second gap **2605** in a spacing direction between the first sidewall **2602** and the second sidewall **2603**. The housing **261** is also referred to as a first housing, and the housing **262** is also referred to as the second housing.

[0117] For example, the housing **262** is provided with the adhesive receiving groove **2601**, and the housing **261** is inserted into the adhesive receiving groove **2601**.

[0118] The housing **261** and the housing **262** may be connected to form a receiving cavity **2600** between the housing **261** and the housing **262**. The receiving cavity **2600** may be configured to accommodate components of the headphone **100**, such as the battery **29**. The battery **29** may be configured to power the headphone **100**.

[0119] In some embodiments, as shown in FIG. **4**, the circuit board **28** may be disposed in the receiving cavity **2600**. The circuit board **28** is configured for converting and processing electrical signals to support the implementation of various functions of the headphone **100**, for example, the circuit board **28** supports the headphone **100** to implement functions such as switching on and off, switching playback contents, increasing and decreasing the sound volume, etc. In some embodiments, the two earhook assemblies **2** include the housing assembly **20** and the housing assembly **26**, one of which is disposed with the battery **29**, and the other is disposed with the circuit board **28**.

[0120] The adhesive may be disposed between the housing **261** and the housing **262**. The adhesive may form a sealing between the housing **262** and the housing **261** to improve the waterproof performance of the headphone **100**. The adhesive may also act as a fixation by adhering. When dispensing adhesive, the adhesive may be in a form of a fluid, and the adhesive may cure after a period of time.

[0121] The adhesive receiving groove **2601** may be configured to hold the adhesive. The dispensing operation of the adhesive is facilitated by setting the adhesive receiving groove **2601** to be exposed externally along the insertion direction of the second of the housing **261** and the housing **262**. By disposing the adhesive receiving groove **2601**, it is convenient to control a dispensing amount when dispensing the adhesive into the adhesive receiving groove **2601**, thereby reducing a risk of the headphone **100** being poorly sealed due to a low dispensing amount of adhesive. At the same time, a probability of the adhesive seeping outwardly onto the outer surfaces of the housing **261** and the housing **262** or seeping inwardly into the receiving cavity **2600** due to too much dispensing of the adhesive may also be reduced, so as to minimize the adverse effect on the operation state and the appearance of the headphone **100**.

[0122] The dispensing of the adhesive may be performed first during the assembly process, and then the second of the housing **261** and the housing **262** may be inserted into the adhesive receiving groove **2601** such that the adhesive fills the first gap **2604** and the second gap **2605**, respectively. By disposing the width L**2** of the first gap **2604** greater than the width L**3** of the second gap **2605** so that the second gap **2605** is able to hold more adhesive, when the second of the housing **261** and the housing **262** is inserted into the adhesive receiving groove **2601**, the adhesive tends to flow into the second gap **2605**, thereby reducing the probability of the adhesive flowing onto the outer surfaces of the housing **261** and the housing **262**, which is conducive to maintaining a good appearance of the headphone **100**.

[0123] In some embodiments, the second of the housing **261** and the housing **262** has a connection boss **2609** inserted within the adhesive receiving groove **2601**.

[0124] As shown in FIGS. **11-13**, a positioning rib **2606** is disposed on the first sidewall **2602** and/or the second sidewall **2603**, and the positioning rib **2606** is configured to define widths of the first gap **2604** and the second gap **2605**. For example, the positioning rib **2606** is disposed on the first sidewall **2602**. In a spaced apart direction of the first sidewall **2602** and the second sidewall **2603**, the positioning rib **2606** may impede the second of the housing **261** and the housing **262** from contacting the first sidewall **2602**. When the second of the housing **261** and the housing **262** abuts against the positioning rib **2606**, a projection height of the positioning rib **2606** may be taken as the width L**2** of the first gap **2604**. When the second of the housing **261** and the housing **262** does not abut the positioning rib **2606**, the width L**2** of the first gap **2604** is greater than the projection height of the positioning rib **2606**. Furthermore, at a design stage of the housing assembly **26**, in the spacing direction of the first sidewall **2602** and the second sidewall **2603**, the width L**2** of the first gap **2604** is set greater than or equal to the positioning rib **2606**, and the width L**3** of the second gap **2605** is set to be less than the projection height of the positioning rib **2606**, such that the width L**2** of the first gap **2604** is greater than the width L**3** of the second gap **2605**.

[0125] As another example, the positioning rib **2606** is disposed on the second sidewall **2603**. In the design stage of the housing assembly **26**, in the spacing direction of the first sidewall **2602** and the second sidewall **2603**, the width L**3** of the second gap **2605** is disposed equal to the projection height of the positioning rib **2606**, and the width L**2** of the first gap **2604** is disposed greater than the projection height of the positioning rib **2606**, such that the width L**2** of the first gap **2604** is greater than the width L**3** of the second gap **2605**. During the assembly process of inserting the second of the housing **261** and the housing **262** into the adhesive receiving groove **2601**, it is necessary to keep the second of the housing **261** and the housing **262** abut against the positioning rib **2606** so that the width L**3** of the second gap **2605** is equal to the projection height of the positioning rib **2606**, such that the width L**3** of the second gap **2605** is less than the width L**2** of the first gap **2604**.

[0126] As another example, the first sidewall **2602** and the second sidewall **2603** are each disposed with the positioning rib **2606**, and in the spacing direction of the first sidewall **2602** and the second sidewall **2603**, the projection height of the positioning rib **2606** on the first sidewall **2602** is greater than the projection height of the positioning rib **2606** on the second sidewall **2603**. During the assembly process of inserting the second of the housing **261** and the housing **262** into the adhesive receiving groove **2601**, the second of the housing **261** and the housing **262** needs to be kept in contact with the positioning rib **2606** on the second sidewall **2603**, so that the width L**2** of the first gap **2604** is greater than the width L**3** of the second gap **2605**.

[0127] In some embodiments, there are two or more spaced apart positioning ribs **2606**.

[0128] In some embodiments, combining FIG. **12** and FIG. **13**, the positioning rib **2606** is disposed on the first sidewall **2602**.

[0129] In some embodiments, combining FIG. **10** and FIG. **11**, a difference between the width L**2** of the first gap **2604** and the width L**3** of the second gap **2605** is greater than or equal to 0.05 mm. For example, the difference between the width L**2** of the first gap **2604** and the width L**3** of the second gap **2605** is 0.06 mm, 0.07 mm, 0.08 mm, 0.09 mm, or 0.1 mm.

[0130] In this way, the amount of adhesive flowing to the second gap **2605** during insertions of the second of the housing **261** and the housing **262** into the adhesive receiving groove **2601** is within an acceptable range, which, in turn, prevents the adhesive from flowing onto the housing **261** and the outer surface of the housing **262**.

[0131] As the height of the different positions of the first gap **2604** relative to the bottom wall of the adhesive receiving groove **2601** changes, the width L**2** of the first gap **2604** may change. As the height of the different positions of the second gap **2605** relative to the bottom wall of the adhesive receiving groove **2601** changes, the width L**3** of the second gap **2605** may change. Therefore, there is a need to compare the width L**2** of the first gap **2604** and the width L**3** of the second gap **2605** at the same height relative to the bottom wall of the adhesive receiving groove **2601** as well as to determine a difference between the width of the first gap **2604** L**2** and the width L**3** of the second gap **2605**.

[0132] As shown in FIG. **10** and FIG. **11**, in the insertion direction of the housing **261** and the housing **262**, the height of the end of the first gap **2604** away from the bottom wall relative to the bottom wall is greater than the height of the end of the second gap **2605** away from the bottom wall relative to the bottom wall.

[0133] The first gap **2604** has a greater width compared to the second gap **2605**, and the adhesive tends to flow into the first gap **2604**. By setting the height of the end of the first gap **2604** away from the bottom wall relative to the bottom wall to be greater than the height of the end of the second gap **2605** away from the bottom wall relative to the bottom wall, it may be possible to make the first gap **2604** capable of holding more adhesive, thereby reducing the probability of the adhesive flowing through the first gap **2604** into the interior of the housing assembly **26**.

[0134] In some embodiments, combining FIG. **10** and FIG. **11**, a height difference H**2** between the end of the first gap **2604** away from the bottom wall and the end of the second gap **2605** away from the bottom wall is greater than or equal to **0.5** mm. For example, the height difference H**2** between the end of the first gap **2604** away from the bottom wall and the end of the second gap **2605** away from the bottom wall is 0.6 mm, 0.7 mm, 0.8 mm, 0.9 mm, or 1 mm.

[0135] In this way, the first gap **2604** is made to hold more adhesive compared to the second gap **2605** during the insertions of the second of the housing **261** and the housing **262** into the adhesive receiving groove **2601** so that more adhesive flows toward the first gap **2064** during squeezing, and the amount of adhesive flowing through the second gap **2604** toward the interior of the housing assembly **26** is within an acceptable range, or the adhesive may not flow toward the interior of the housing assembly **26**.

[0136] As shown in FIG. **10** and FIG. **11**, the adhesive receiving groove **2601** includes a third sidewall **2607** connected to the first sidewall **2602**, and the third sidewall **2607** is far from the bottom wall of the adhesive receiving groove **2601** as compared to the first sidewall **2602**. A third gap **2608** is formed between the third sidewall **2607** and the second of the housing **261** and the housing **262**, with a width of the third gap **2608** gradually increasing in a spacing direction away from the bottom wall. For example, the third sidewall **2607** is disposed in the housing **262** and forms a third gap **2608** between the housing **262** and the housing **261**.

[0137] The third gap **2608** is configured to accommodate the adhesive overflow from the second gap **2605** when more adhesive flows into the second gap **2605** and overflows from the second gap **2605**. By disposing the third gap **2608**, the risk of adhesive flowing from the second gap **2605** onto the outer surfaces of the housing **261** and the housing **262** may be reduced.

[0138] In the spacing direction, by setting the width of the third gap **2608** to gradually increase in the direction away from the bottom wall of the adhesive receiving groove **2601**, the amount of adhesive held by the third gap **2608** may be increased, thereby further reducing the risk of adhesive flowing onto the housing **261** and the outer surface of the housing **262**.

[0139] In some embodiments, the third sidewall **2607** may also guide during the insertions of the second of the housing **261** and the housing **262** into the adhesive receiving groove **2601**, thereby facilitating the insertions of the second of the housing **261** and the housing **262** into the adhesive receiving groove **2601** and improving the assembly efficiency.

[0140] A portion of the third gap **2608** may have a width greater than the width L**2** of the first gap **2604**, as shown in FIG. **10** and FIG. **11**.

[0141] The width of the portion of the third gap **2608** near the bottom wall of the adhesive receiving groove **2601** is smaller, thereby reducing the amount of adhesive that the third gap **2608** is able to hold. By setting the width of the portion of the third gap **2608** that is away from the bottom wall of the adhesive receiving groove **2601** to be greater than the width L**2** of the first gap **2604**, the amount of the adhesive held by the third gap **2608** may be increased, thereby reducing the risk of the adhesive flowing onto the outer surfaces of the housing **261** and the housing **262**.

[0142] As shown in FIG. **10** and FIG. **11**, a height of the end of the first gap **2604** away from the bottom wall of the adhesive receiving groove **2601** relative to the bottom wall of the adhesive receiving groove **2601** is greater than a height of the end of the third gap **2608** away from the bottom wall of the adhesive receiving groove **2601** relative to the bottom wall of the adhesive receiving groove **2601**, i.e., there is a height difference H**3**.

[0143] By setting the height of the end of the first gap **2604** away from the bottom wall of the adhesive receiving groove **2601** relative to the bottom wall of the adhesive receiving groove **2601** to be greater than the height of the end of the third gap **2608** away from the bottom wall of the adhesive receiving groove **2601** relative to the bottom wall of the adhesive receiving groove **2601**, it is possible to enable the first gap **2604** to hold more adhesive, thereby reducing the probability of the adhesive flowing to the interior of the housing assembly **26**.

[0144] On the other hand, the first sidewall **2602** may act as a limit. Specifically, the third gap **2608** may act as a guide, and during the insertions of the second of the housing **261** and the housing **262** into the adhesive receiving groove **2601**, the second of the housing **261** and the housing **262** moves toward the first sidewall **2602**. By setting the height of the end of the first gap **2604** away from the bottom wall relative to the bottom wall to be greater than the height of the end of the third gap **2608** away from the bottom wall relative to the bottom wall, when the second in the housing **261** and the housing **262** move toward the first sidewall **2602**, the first side wall **2602** may play a limiting role, so that the second in the housing **261** and the housing **262** is inserted smoothly into the adhesive receiving groove **2601**, and the assembly efficiency may be improved.

[0145] As shown in FIG. **10** and FIG. **11**, the housing assembly **26** may include an elastic wrapping layer **263** partially wrapped on an outer surface of the housing **261**, the adhesive receiving groove **2601** is disposed in the housing **262**, and an exposed portion of the housing **261** that is not wrapped by the elastic wrapping layer **263** is inserted into the adhesive receiving groove **2601**.

[0146] Specifically, the exposed portion of the housing **261** not wrapped by the elastic wrapping layer **263** may be the connection boss **2609**.

[0147] By disposing the elastic wrapping layer **263** on the outer surface of the housing **261**, the comfort of the user when holding the headphone **100** is improved, and the friction is increased.

[0148] By inserting the exposed portion of the housing **261** into the adhesive receiving groove **2601**, the housing **261** may position the housing **262**, which limits the relative movement of the housing **261** and the housing **262**, and increases a stability of the connection between the housing

261 and the housing **262**.

[0149] By setting the exposed portion of the housing **261** to be wrapped by the elastic wrapping layer **263**, an interference of the elastic wrapping layer **263** on the connection between the housing **261** and the housing **262** within the adhesive receiving groove **2601** may be avoided.

[0150] In some embodiments, the housing **262** is abut against the elastic wrapping layer **263** along the insertion direction of the exposed portion relative to the housing **262**, which makes the outer surface of the housing **262** and the outer surface of the elastic wrapping layer **263** to be flush at an offset point, which on the one hand makes the housing **262** and the elastic wrapping layer **263** closer at the offset point, provides a better sealing effect, reduces the entry of the debris and the water vapor, and improves a service life of the elastic wrapping layer **263**, and on the other hand, the flush at the offset reduces the discomfort to the user when being touched, which makes the headphone **100** more integrative and more comfortable, and also more aesthetically pleasing.

[0151] In some embodiments, the housing **262** is not wrapped by the elastic wrapping layer **263**, and an outer surface area of the housing **262** may be smaller than an outer surface area of the housing **261**, and the user tends to hold the housing **261** to hold the headphone **100**, such that the outer surface of the housing **262** may not be disposed with the elastic wrapping layer **263**.

[0152] As shown in FIG. **14** and FIG. **15**, the earhook assembly **2** may include the housing assembly **20**, an interface assembly **22**, a stopper **25**, the circuit board **28**, and the lead assembly **24**. In some embodiments, the headphone **100** may also include a wearing assembly **30** and the dust plug **23**.

[0153] In some embodiments, as shown in FIGS. **15-18**, the housing assembly **20** includes the housing **201** disposed with the accommodation groove **2004** and the receiving cavity **2010** connected to the accommodation groove **2004**. The interface assembly **22** is inserted into the accommodation groove **2004**. The circuit board **28** is disposed within the receiving cavity **2010**. The lead assembly **24** electrically connects the circuit board **28** to the interface assembly **22**.

[0154] By disposing the interface assembly **22** and the circuit board **28** separately instead of fixing and soldering the interface assembly **22** directly to the circuit board **28**, the lead assembly **24** is configured to connect the interface assembly **22** and the circuit board **28**. In this way, a degree of freedom of a mounting position and a mounting attitude of the interface assembly **22** is improved. On this basis, the accommodation groove **2004** and the receiving cavity **2010** connected with the accommodation groove **2004** are reasonably disposed inside the housing **201** for holding the interface assembly **22** and the circuit board **28**, respectively, which is favorable to a flexible application of the space inside the housing **201** to mount the interface assembly **22** and the circuit board **28** respectively, thereby facilitating the assembly of the headphone **100**.

[0155] As shown in FIG. **16**, the circuit board **28** is spaced apart from the interface assembly **22** in an insertion direction f**2** of the interface assembly **22** relative to the accommodation groove **2004**.

[0156] By maintaining a certain spacing between the interface assembly **22** and the circuit board **28** in the insertion direction f**2** of the interface assembly **22** relative to the accommodation groove **2004**, the interface assembly **22** effectively reduces an encroachment on a space of the receiving cavity **2010** where the circuit board **28** is located, which effectively reduces a possibility of the interface assembly **22** interfering with the circuit board **28**, improves an effectiveness of the operation of the interface assembly **22** and the circuit board **28**, and improves the reliability and the service life of the headphone **100**.

[0157] In some embodiments, as shown in FIGS. **16-19**, the housing **201** is disposed with an insert groove **2001** spaced apart from the accommodation groove **2004** by an interval wall **2018**. The interval wall **2018** is disposed with a connection groove **2019** connecting the insert groove **2001** and the accommodation groove **2004**. The wearing assembly **30** includes an insert member **33**, the insert member **33** comprising an insert body **331** inserted within the insert groove **2001**. The stopper **25** is disposed to be able to be inserted into the insert groove **2001** from the accommodation groove **2004** through the connection groove **2019** and is configured for snapping the insert body **331** into the insert groove **2001**. The interface assembly **22** is inserted into the accommodation groove **2004**.

[0158] By disposing the interval wall **2018** to form the insert groove **2001** and the accommodation groove **2004** isolated from each other, and configured for the insertion of the insert member **33** and the interface assembly **22**, respectively, while the insert member **33** and the interface assembly **22** are effectively limited and mounted, the insert member **33** and the interface assembly **22** do not interfere with each other, and do not encroach on the space of each other, which are convenient for assemble and disassemble, and are conducive to a maintenance and care of the headphone **100**. The connection groove **2019** disposed on the interval wall **2018** is configured for the insertion of the stopper **25** to stop the insert body **331** into the insert groove **2001**, so that the stopper **25** limits the insert body **331** when not exposed outside the housing **201**. In this way, the stability and reliability of the mounting of the insert body **331** is improved without affecting the appearance of the headphone **100**, which effectively improves the aesthetics of the headphone **100**. In addition, disposing the connection groove **2019** in the interval wall **2018** may improve a utilization rate of the space, realize a close arrangement inside the housing assembly **20**, and reduce the volume of the headphone **100**.

[0159] In some embodiments, a rear end portion of the housing assembly **20** (i.e., an end portion disposed with a groove opening of the insert groove **2001** and a groove opening of the accommodation groove **2004**, which away from the ear in the wearing state) is inclined, which, on the one hand, is designed to reduce the volume and a weight of the entire housing assembly **2** and make the entire headphone **100** miniaturized, and on the other hand, makes it easy for mold opening and demolding, thereby reducing a manufacturing difficulty, and making the headphone **100** more aesthetically pleasing. By setting the interface assembly **22** at the inclined rear end portion, a slope region which is not easy to be utilized may be efficiently utilized, and an overall structure may be more compact, and the space utilization may be high. Additionally, the interface assembly **22** is disposed at the rear end portion instead of a lower side surface, which has less impact on the stability of a placement of the headphone when in use. Additionally, cables connecting to the interface assembly **22** are less likely to wobble with the interface assembly **22** and rub unnecessarily against each other, and the cables are less likely to rub against the region where they are placed.

[0160] In some embodiments, the interface assembly **22** and the housing **202** are fixed by dispensing adhesive to achieve a fixation between the interface assembly **22** and the housing **202** as well as a fixation and sealing of a connection position between the interface assembly **22** and the housing **202**. In some embodiments, the interface assembly **22** and the housing **202** are fixed by disposing a silicone member (not shown in the figure). The silicone member is interference fit with the interface assembly **22** and the housing **202**, and the interface assembly **22** and the housing **202** are fixed through snap screws and other fixing structures, so as to realize the fixing between the interface assembly **22** and the housing **202** as well as the fixing and sealing between the interface assembly **22** and the connecting position of the housing **202**.

[0161] In some embodiments, as shown in FIG. **18**, the position of the connection groove **2019** is set such that the stopper **25** is visible from an exterior of the housing **201** in an insertion direction f**3** along the stopper **25** relative to the connection groove **2019**.

[0162] The connection groove **2019** is visible from the exterior of the housing **201**, i.e., the position of the connection groove **2019** is clearly exposed to the exterior of the housing **201**, so that when inserted along the insertion direction f**3** of the stopper **25** relative to the connection groove **2019**, the stopper **25** has a clear position, and is able to operate easily, so as to effectively reduce the assembly difficulty and facilitate assembly and disassembly.

[0163] In some embodiments, as shown in FIGS. **16-18**, the housing assembly **20** further includes the housing **202**, which is connected to the housing **201** and is configured to shield the connection groove **2019** and the stopper **25**.

[0164] The connection between the housing **202** and the housing **201** may shield the connection groove **2019** and the stopper **25** to make the connection groove **2019** and the stopper **25** invisible from the outside of the headphone **100**, thereby improving the aesthetics of the headphone **100**. Compared to an integrated housing, the provision of the split-connected housing **201** and the housing **202** allows the mounting of the stopper **25** when the housing **202** is not connected to the housing **201**, which is easy to operate and less difficult to install. After inserting the stopper **25**, by connecting the housing **202** and the housing **201**, the connection groove **2019** and the stopper **25** may be shielded, which ensures that the headphone **100** assembly is operable and effectively safeguards the aesthetics of the headphone **100**.
[0165] In some embodiments, as shown in FIG. **16** and FIG. **19**, the housing **202** is further disposed to be capable of stopping the stopper **25** in an opposite direction of the insertion direction f**3** of the stopper **25** relative to the connection groove **2019**.
[0166] The housing **202**, while shielding the connection groove **2019** and the stopper **25**, is able to stop the stopper **25** to support the stopper **25**, thereby avoiding that the stopper **25** from moving in the opposite direction of the direction f**3**, which causes the insert member **33** to loosen and affect the use of the headphone **100**. In this way, the stability of the connection is enhanced, and the service life of the headphone **100** is improved.
[0167] In some embodiments, as shown in FIG. **18** and FIG. **19**, the interface assembly **22** includes a main body portion **221** and a wire arrangement portion **222**. The main body portion **221** is provided with a first guiding cavity **2210**, and the wire arrangement portion **222** is disposed within the first guiding cavity **2210**. The housing **202** is disposed with a second guiding cavity **2022**, the housing **202** is connected to the housing **201**, and the second guiding cavity **2022** is connected to the first guiding cavity **2210**.
[0168] By connecting the housing **202** to the housing **201** to enable the first guiding cavity **2210** and the second guiding cavity **2022** to connect to each other, the interface assembly **22** does not need to set up a complete guiding cavity for the wire arrangement portion **222**, and the second guiding cavity **2022** is constructed using the housing **202**, which reduces the overall volume of the housing assembly **20** and makes the housing assembly **20** more compact.
[0169] The wire arrangement portion **222** protrudes from the first guiding cavity **2210** and extends into the second guiding cavity **2022**. In the above way, the first guiding cavity **2210** and the second guiding cavity **2022** are made to collectively accommodate the wire arrangement portion **222** to facilitate a plug of the wire arrangement portion **222**.
[0170] As shown in FIG. **19**, the interface assembly **22** may further include a flange portion **223** disposed on a side of the main body portion **221** away from the receiving cavity **2010**. The housing **201** is disposed with a first support platform **2015**, the flange portion **223** is supported on the first support platform **2015**, and the housing **202** is disposed to shield the flange portion **223**.
[0171] By disposing the flange portion **223** supported on the first support platform **2015**, the support of the flange portion **223** by the first support platform **2015** may improve the stability and reliability of the mounting of the interface assembly **22**, so as to effectively limit and mount the interface assembly **22**. Additionally, the housing **202** is disposed to shield the flange portion **223**, and the flange portion **223** is not exposed to the exterior of the housing assembly **20**, so that the flange portion **223** is invisible by the user from the exterior of the headphone **100**, which effectively enhances the aesthetics of the headphone **100**.
[0172] As shown in FIG. **19**, the housing **202** may further press the flange portion **223** against the first support platform **2015**.
[0173] The flange portion **223** is pressed by the housing **202** against the first support platform **2015**, thereby making the flange portion **223** to be tightly supported against the first support platform **2015**, improving a support effect of the first support platform **2015** on the flange portion **223**, and further making the housing assembly **20** more stable and reliable for the limiting and mounting of the interface assembly **22**. In this way, the possibility of a shaking of the interface

assembly **22** is reduced, and the fixing effect is effectively improved.
[0174] As shown in FIG. **19**, the first support platform **2015** is disposed with an adhesive receiving groove **2016**, and the flange portion **223** covers the adhesive receiving groove **2016**.
[0175] The adhesive receiving groove **2016** may hold the adhesive, so that when the interface assembly **22** is mounted, the adhesive held in the adhesive receiving groove **2016** may fill a gap between the interface assembly **22** and the housing **201**, thereby improving the sealing. The use of the sealant may effectively fix the interface assembly **22**, and increase the stability of the connection between the interface assembly **22** and the housing **201**.
[0176] As shown in FIG. **18** and FIG. **19**, the first support platform **2015**, the flange portion **223**, and the adhesive receiving groove **2016** are set to be closely surround the accommodation groove **2004** around the interface assembly **22** relative to the insertion direction f**2** of the accommodation groove **2004**.
[0177] The first support platform **2015**, the flange portion **223**, and the adhesive receiving groove **2016** are all closely surround the containment groove **2004** around the interface assembly **22**, i.e., the flange portion **223** is supported on the first support platform **2015** around the interface assembly **22**, which effectively improves a contact area between the first support platform **2015** and the flange portion **223**, and makes a support force provided by the first support platform **2015** to the interface assembly **22** uniform and reliable, thereby effectively improving the support effect of the first support platform **2015** on the interface assembly **22**, improving the fixing effect, and improving the service life of the headphone **100**. Additionally, the adhesive receiving groove **2016** that closely surrounds the accommodation groove **2004** around the interface assembly **22** allows for a complete dispensing of the adhesive around the accommodation groove **2004** when the interface assembly **22** is installed, thereby effectively improving the sealing effect.
[0178] In some embodiments, combining FIG. **18** and FIG. **19**, the housing **201** is further disposed with a second support platform **2017**, and an inner end surface of a side of the main body portion **221** towards the receiving cavity **2010** is supported on the second support platform **2017**.
[0179] By setting the second support platform **2017** to support the inner end surface of the side of the main body portion **221** towards the receiving cavity **2010**, the first support platform **2015** and the second support platform **2017** jointly support the interface assembly **22** to improve the stability of the interface assembly **22**, and reduce the possibility of shaking of the interface assembly **22**.
[0180] In some embodiments, as shown in FIG. **20** and FIG. **21**, the stopper **25** includes two insert pins **251** disposed side-by-side and a connection portion **252** connecting the two insert pins **251**. The insert body **331** is disposed with two slots **335**, the two pins **251** are respectively inserted into the two slots **335**, and the connection portion **252** is disposed in the connection groove **2019** and forms an interference with a groove wall of the connection groove **2019** at least in the opposite direction of the insertion direction f**1**.
[0181] The stopper **25** is configured to stop and limit the insert body **331** to realize the limiting and mounting of the insert member **33**, and by disposing two pins **251** side by side in the stopper **25**, and by making the insert body **331** disposed with two slots **335** that cooperate with the two pins **251** to connect the stopper **25** and the insert body **331**, the two pins **251** may be correspondingly inserted in the two slots **335**, and the two pins **251** may improve the connection stability and improve the effect of the stop.
[0182] The connection portion **252** is disposed in the connection groove **2019**, and the connection portion **252** forms the interference with the groove wall of the connection groove **2019** at least in the opposite direction of the insertion direction f**1**, so as to make the slot wall of the connection groove **2019** provide a supporting force to the connection portion **252**, which in turn reduces an overall height of the stopper **25** and improves the stability and reliability of the connection.
[0183] As shown in FIG. **19** and FIG. **21**, the connection portion **252** may be sunk in the connection groove **2019**, and the housing **202** may be disposed with an adhesive receiving groove **2023** connected to the connection groove **2019**.

[0184] The adhesive receiving groove **2023** may hold the sealant, so that when the housing **202** is mounted, the sealant held in the adhesive receiving groove **2023** may fill the gap between, for example, the housing **202** and the connection groove **2019**, and the gap between the housing **202** and the connection portion **252**, to improve the sealing. The use of the sealant effectively fixes the stopper **25** and the housing **202**, increasing the stability of the connection between the housing **201**, the housing **202**, and the stopper **25**.
[0185] In some embodiments, as shown in FIG. **19**, the interval wall **2018** is further disposed with the first support platform **2015**. The flange portion **223** is supported on the first support platform **2015**.
[0186] By disposing the flange portion **223** and the support platform **2015**, and by supporting the flange portion **223** on the first support platform **2015**, the support of the flange portion **223** by the support platform **2015** is able to improve the stability and reliability of the mounting of the interface assembly **22**, and effectively limiting and mounting the interface assembly **22**.
[0187] As shown in FIG. **19**, the flange portion **223** may be further disposed to be capable of stopping the stopper **25** in the opposite direction of the insertion direction f**3**.
[0188] By disposing the flange portion **223** to stop the stopper **25** in the opposite direction of the insertion direction f**3**, a supporting force is further provided for the stopper **25** to stop and limit the stopper **25**. In this way, the structure of the headphone **100** is made more stable and solid, the possibility of shaking and loosening of the stopper **25** is reduced, and the service life of the headphone **100** is increased.
[0189] In some embodiments, as shown in FIG. **16**, the insertion direction f**2** is set at an acute angle to the insertion direction f**1**. Specifically, the insertion direction f**1** is disposed at an acute angle to the insertion direction f**2**. The second support platform **2017** is disposed on a side of the interface assembly **22** away from the insert member **33**, and an inner end surface of the main body portion **221** is suspended on the side of the body portion **221** towards the insert member **33**.
[0190] On the basis that the insertion direction f**2** is set at the acute angle to the insertion direction f**1**, a side of an inner end surface of the main body portion **221** is supported on the second support platform **2017**, and a side towards the insert member **33** is suspended, which reduces the volume of the housing assembly **20** and improves the space utilization inside the housing assembly **20** while effectively supporting and fixing the interface assembly **22**.
[0191] In some embodiments, as shown in FIG. **16**, a side of the interface assembly **22** towards the interior of the housing **201** is disposed to exceed the interval wall **2018** in the insertion direction f**2**.
[0192] On the basis that the insertion direction f**2** is set at the acute angle to the insertion direction f**1**, the interface assembly **22** exceeds the interval wall **2018**, so as to further make full use of the space inside the housing assembly **20** without interference between the interface assembly **22** and the insert body **331**, and to improve the space utilization inside the housing assembly **20**, and make the structure inside the housing assembly **20** more compact, improve an integrality and a tightness of the structure, reduce volumes of the housing assembly **20** and the headphone **100**, make the headphone **100** more convenient to wear and carry, and enhance a sense of use of the headphone **100**.
[0193] The interface assembly **22** is inserted in the accommodation groove **2004** as shown in FIGS. **22-24**. The interface assembly **22** is configured for a corresponding external interface to be plugged in to enable the headphone **100** to electrically connect to other devices for functions such as charging, transmission, etc. The dust plug **23** may be partially fixed to the housing assembly **20** through the cooperation between the housing **201** and the housing **202**, and may selectively shield the interface assembly **22**.
[0194] In some embodiments, the wearing assembly **30** may include the insert member **33**. The insert member **33** may be inserted into the housing assembly **20** to enable the wearing assembly **30** to be fixed to the housing **201**. The wearing assembly **30** is configured to be worn on an auricle, the neck, the back of the head, etc., to fix the headphone **100** at a certain position.

[0195] In some embodiments, as shown in FIGS. **23** to FIG. **24**, the headphone **100** further includes the key assembly **21** disposed within the housing **201** and exposed through the housing **202**, and specifically, the key assembly **21** is configured to be operated by the user for realizing a human-machine interaction function between the headphone **100** and the user.
[0196] Specifically, as shown in FIG. **23** and FIG. **24**, the dust plug **23** may include the plug body **231** and the fixing portion **232**, and the fixing portion **232** may be limited to the housing **201** by the housing **202**. The plug body **231** is connected to the fixing portion **232** and is configured to selectively shield the interface assembly **22**.
[0197] The plug body **231** is capable of moving relative to the interface assembly **22** so as to selectively shield or expose the interface assembly **22**. The fixing portion **232** connects the plug body **231** to pull the plug body **231**, such that the plug body **231** is still connected to the housing assembly **20** through the fixing portion **232** when exposing the interface assembly **22**, and thus is less likely to fall off from the headphone **100**.
[0198] By disposing the dust plug **23** to selectively cover the interface assembly **22**, the interface assembly **22** may not be shielded by the dust plug **23** when the interface assembly **22** is not connected to the corresponding plug for use, so as to reduce a situation where impurities, such as dust particles, fall into the interface assembly **22** and block the interface assembly **22** and affect the use of the interface assembly **22**. The fixing portion **232** of the dust plug **23** is limited to the housing **201** through the housing **202**, and the housing **201** and the housing **202** are configured to fix the dust plug **23** together to make the assembly of the dust plug **23** more stable, or the fixing portion **232** may not be directly fixed on the housing **201**, so that no corresponding fixed position is disposed on the shell **201**, and the structures of the housing **201** and the housing **202**, as well as an assembly manner of the dust plug **23** are simplified.
[0199] As shown in FIGS. **23-28**, the main body portion **221** may be provided with the first guiding cavity **2210**, the wire arrangement portion **222** is disposed in the first guiding cavity **2210**, the housing **202** is disposed with the second guiding cavity **2022**. The second guiding cavity **2022** is connected to the first guiding cavity **2210**, and the plug body **231** is disposed to be able to selectively insert into or remove from the first guiding cavity **2210**.
[0200] Specifically, the main body portion **221** is configured for plugging an external interface corresponding to the interface assembly **22**. The wire arrangement **222** protrudes from the first guiding cavity **2210** and extends into the second guiding cavity **2022**. After the external interface is plugged in, the wire arrangement portion **222** is configured to enable the headphone **100** to be electrically connected to the external device through the wire arrangement portion **222**. In the above manner, the first guiding cavity **2210** and the second guiding cavity **2022** are made to collectively accommodate the wire arrangement portion **222**, which enables the wire arrangement portion **222** to be conveniently plugged.
[0201] In some embodiments, the plug body **231** may be inserted into the first guiding cavity **2210** through the second guiding cavity **2022** to shield the interface assembly **22**. A shape of the plug body **231** may be adapted to the first guiding cavity **2210** and the wire arrangement portion **222** such that the wire arrangement portion **222** is less likely to obstruct the plug body **231** from being inserted into the first guiding cavity **2210**. By disposing the second guiding cavity **2022** to be connected to the first guiding cavity **2210**, the interface assembly **22** may not have a complete guiding cavity. On the one hand, is able to the volume of the interface assembly **22** is reduced, and the volume of the entire housing assembly **20** is further reduced.
[0202] Through the connection between the second guiding cavity **2022** and the first guiding cavity **2210**, the interface assembly **22** may be pressed between the housing **201** and the housing **202**, which facilitates the mounting of the interface assembly **22**. An inserted portion of the plug body **231** is set to be able to fit into the second guiding cavity **2022** and the first guiding cavity **2210**, so that the plug body **231**, even after being inserted into the first guiding cavity **2210**, is also able to be fixed relative to the interface assembly **22** instead of easily falling off from the first guiding cavity

2210 and the second guiding cavity **2022**.

[0203] In some embodiments, when inserted into the first guiding cavity **2210**, a side of the plug body **231** away from the interface assembly **22** may completely cover the second guiding cavity **2022** to completely shield the interface assembly **22**, making it less likely for the impurities such as dust, metal particles, etc. to fall into the interface assembly **22**.

[0204] In some embodiments, the plug body **231** may be an elastomer made of an elastic material such as the silicone, the rubber, etc., so as to allow the plug body **231** to selectively inserted into or removed out of the first guiding cavity **2210**. During the process, the plug body **231** is less prone to damage the Interface assembly **22**.

[0205] By setting the second guiding cavity **2022** and the first guiding cavity **2210**, it is easy for the plug body **231** to be selectively inserted into or removed out of the first guiding cavity **2210** to allow the plug body **231** to selectively shield the interface assembly **22**, thereby reducing a situation where the impurities such as the dust, the metal particles, etc. falling into the interface assembly **22**.

[0206] In some embodiments, as shown in FIG. **23-28**, the insert member **33** is inserted within the insert groove **2001**, and the fixing portion **232** is located on a side of the interface assembly **22** away from the insert member **33**. Specifically, as shown in FIG. **24**, the fixing portion **232** and the insert member **33** are located on both sides of the interface assembly **22**, so that the positions of the dust plug **23** and the insert member **33** do not interact with each other, the space of the headphone **100** is saved, and the space utilization rate of the headphone **100** is improved.

[0207] In some embodiments, the fixing portion **232** is disposed between the key assembly **21** and the interface assembly **22** to save a using space of the headphone **100** reduce the volume of the headphone **100**, and improve the space utilization of the headphone **100**.

[0208] In some implementations, as shown in FIGS. **24-29**, the housing **202** is connected to the housing **201**, and the fixing portion **232** is pressed and limited to the housing **201**. Specifically, the housing **202** is capped over the housing **201**, and the fixing portion **23** is partially disposed in the housing **201** and is limited to the housing **201** while being pressed by the housing **202**. In this way, the assembly of the dust plug **23** is more convenient, and the headphone **100** is easily disassembled and maintained.

[0209] In some embodiments, as shown in FIG. **25**, a positioning hole **235** may be disposed on the first of the fixing portion **232** and the housing **201**, and a positioning post **236** may be disposed on the second of the fixing portion **232** and the housing **201**. The positioning post **236** is inserted within the positioning hole **235**, and the housing **202** presses and limits the fixed portion **232** on the housing **201** at a periphery of the positioning hole **235**.

[0210] By disposing the positioning holes **235** and the positioning posts **236**, the fixing portion **232** is connected to the housing **201**, and then the housing **202** is configured to press and limit the positioning post **236** to the positioning holes **235** at the periphery of the positioning holes **235**, then the fixing portion **232** may be fixed on the housing **201**. Such setting facilitates the assembly and disassembly of the fixing portion **232**, the housing **201**, and the housing **202**, and at the same time makes the structure of the headphone **100** more compact, and the fixing portion **232** may be more stably limited to the housing **201**.

[0211] In some embodiments, the fixing portion **232** may be the elastomer, the positioning hole **235** may be disposed on the fixing portion **232**, the positioning post **236** may be disposed on the housing **201**, and the positioning post **236** may be inserted relative to the positioning hole **235** at a depth smaller than a hole depth of the positioning hole **235**. Specifically, positions of the positioning hole **235** on the fixing portion **232** and the positioning post **236** on the housing **201** may correspond to each other, the positioning post **236** is inserted in the positioning hole **235**, and an extension direction of the positioning post **236** may be as shown by the I arrow in FIG. **25**. The housing **202** may be pressed toward a direction of the positioning post **236** in the extension direction of the positioning post **236**, so that the fixing portion **232** is fixed to the housing **201** after the housing **201** is connected and fixed to the housing **202**.

[0212] The fixing portion **232** may be the elastomer such as the silicone, the rubber, etc. By setting the insertion depth of the positioning holes **235** less than the hole depth of the positioning holes **235**, the fixing portion **232** may have a certain compression space in the extension direction of the positioning post **236**, and the fixing portion **232** may undergo an elastic deformation when pressed by the housing **202** and the housing **201**, and thus is fixed and limited by the housing **201** and the housing **202**, so that the fixing portion **232** is more securely limited to the housing **201** and is not prone to loosening, thereby improving a structural tightness and solidity of the headphone **100**.

[0213] In some other embodiments, the housing **202** is connected to the housing **201**, and the fixing portion **232** is connected to the housing **202**, as shown in FIGS. **26-27**. The housing **202** and the housing **201** are fixedly covered with each other, by connecting the fixing portion **232** to the housing **202**, the fixing portion **232** is limited to the housing **201**.

[0214] In some embodiments, as shown in FIG. **27**, a mounting hole **2021** may be disposed in the housing **202**, and the dustproof plug **23** may further comprise a connection post **233** connected to the fixing portion **232**, and a stopper **234** connected to the connection post **233**, the connection post **233** is inserted in the mounting hole **2021**, the fixing portion **232** and the stopper **234** are stopped on both sides respectively. The connection post **233** is inserted in the mounting hole **2021**, and the fixing portion **232** and the stopper **234** are clamped to the two sides of the housing **202** respectively.

[0215] Specifically, one end of the fixing portion **232** is connected to the plug body **231**, and one end far from the plug body **231** is connected to the connection post **233**. The fixing portion **232** and the stopper **234** are respectively disposed at the two ends of the connection post **233**, and the plug body **231** is fixed to the housing **202** through the fixing portion **232** and the stopper **234**. The stopper **234** is disposed in a gap between the housing **202** and the housing **201**, and is capable of being blocked by the housing **202** and fixed relative to the housing **202** and the housing **201**, so that the stopper **234** is less likely to move in the direction of the housing **202** and fall off from the housing **202**.

[0216] In some embodiments, the stopper **234** is the elastomer and is disposed so as to be able to pass through the mounting hole **2021** under an action of an external traction force. The stopper **234** may be the elastomer such as the silicone, the rubber, etc., which is capable of elastically deforming under the external force and recovering to an original state when the external force is removed.

[0217] Based on the foregoing description, the dust plug **23** may further include a traction portion **237**. The traction portion **237** is connected to an end of the stopper **234** away from the connection post **233**. The traction portion **237** is configured to be pulled to drive the stopper **234** through the mounting hole **2021** under the action of the external traction force, thereby completing the assembly of the fixing portion **232**. And, after passing through the mounting hole **2021**, the fixing portion **234** may be compressed in the gap between the housing **202** and the housing **201**, so that the mounting hole **2021** is fixed relative to the housing **202** and the housing **201**, and it is not easy to rotate, loosening, or disengage from the mounting hole **2021** so as to dislodge the dust plug **23**. After the stopper **234** passes through the mounting hole **2021** and is fixed, the traction portion **237** may be removed, for example, sheared off, which saves the internal space of the housing **201** and improves the space utilization of the headphone **100**.

[0218] By setting the stopper **234** as the elastomer, the dust plug **23** is made easy to assemble to simplify an assembling process of the headphone **100**, and it further facilitates the disassembly of the dust plug **23** for maintenance, and enables the connection structure of the headphone **100** more tight and stabilized.

[0219] As shown in FIG. **28** to FIG. **31**, the housing assembly **10** is provided with an insert groove **1008**. The wearing assembly **27** includes a lead assembly **271** and an insert member **272**. The insert member **272** includes an insert body **2721** that is insertable within the insert groove **1008**. The lead assembly **271** is threaded within the insert body **2721**, and has an exposed portion **2711** that is externally exposed from an inner end surface of a side of the insert body **2721** towards the interior of the housing assembly **10**. The insert member **272** further includes an annular rib **2722** protruding from the inner end surface, and the annular rib **2722** is disposed around the exposed portion **2711** so that an inner annular surface of the annular rib **2722** forms a first adhesive receiving groove **2723** at a periphery of the exposed portion **2711**. The annular rib **2722** and the insert body **2721** may be integrally molded. The annular rib **2722** may also be considered as a portion of the insert body **2721**.

[0220] A waterproof seal is achieved by filling the first adhesive receiving groove **2723** with adhesive, which in turn seals an assembly gap between the lead assembly **271** and the insert body **2721**. Furthermore, as the first adhesive receiving groove **2723** is formed by the inner annular surface of the annular rib **2722** at the periphery of the exposed portion **2711** of the lead assembly **271**, the lead assembly **271** and the insert body **2721** are more tightly connected, thereby improving the connection stability between the lead assembly **271** and the insert body **2721**.

[0221] Referring to FIGS. **28-33**, in some embodiments, an outer annular surface of the annular rib **2722** cooperates with the groove wall to form a second adhesive receiving groove **2724**. The second adhesive receiving groove **2724** holds the adhesive for the assembly gap between the housing assembly **10** and the insert body **2721** for a corresponding waterproof seal. In some embodiments, by filling the second adhesive receiving groove **2724** with the adhesive, the housing assembly **10** may be connected to the inset body **2721** more tightly, thereby preventing the inset body **2721** from wobbling in the housing assembly **10**.

[0222] In some embodiments, the second adhesive receiving groove **2724** is disposed in an annular shape around the periphery of the annular rib **2722**, allowing the sealant to achieve an annular and more comprehensive sealing around the periphery of the annular rib **2722** to reduce an entry of the water vapor etc., into the housing assembly **10** through the groove gap between the insert body **2721** and the groove wall of the insert groove **1008**.

[0223] Referring to FIGS. **28-32**, in some embodiments, a depth of the first adhesive receiving groove **2723** is greater than a depth of the second adhesive receiving groove **2724** in the insertion direction of the insert member **272** relative to the insert groove **1008**, so as to enable the first adhesive receiving groove **2723** to hold more adhesive, which in turn improves a fixing effect and a sealing effect on the lead assembly **271**. The shallower second adhesive receiving groove **2724** may enable the insert body **2721** to have more region connecting to the insert groove **1008**, which improves an insert fixing effect, and the adhesive within the second adhesive receiving groove **2724** may also provide a good sealing effect.

[0224] In some embodiments, as shown in FIG. **31**, the outer annular surface of the annular rib **2722** includes a first outer annular sub-surface **2725**, a connection surface **2726**, and a second outer annular sub-surface **2728**. The first outer annular sub-surface **2725** is closer to the interior of the housing assembly **10** as compared to the second outer annular sub-surface **2728**, a gap between the first outer annular sub-surface **2725** and the groove wall is greater than a gap between the second outer annular sub-surface **2728** and the groove wall. The connection surface **2726** connects the first outer annular sub-surface **2725** and the second outer annular sub-surface **2728**, and the first outer annular sub-surface **2725**, the connection surface **2726**, and the groove wall of the insert groove **1008** fit together to form the second adhesive receiving groove **2724**.

[0225] Referring to FIGS. **28-33**, in some embodiments, in a direction perpendicular to the insertion direction of the insert member **272** relative to the insert groove **1008**, a width H**4** of the first adhesive receiving groove **2723** is greater than a width H**5** of the second adhesive receiving groove **2724**. The width H**5** shown in FIG. **33** is labeled on the annular rib **2722** only, but should be understood to be a distance between the first outer annular sub-surface **2725** of the annular rib **2722** and a corresponding groove wall of the insert groove **1008**. In this way, on the one hand, the first adhesive receiving groove **2723** may hold more adhesive, which in turn improves the fixing effect and the sealing effect on the lead assembly **271**, and on the other hand, the sealing of the second adhesive receiving groove **2724** may be satisfied with less adhesive.

[0226] Furthermore, as the width H**5** of the second adhesive receiving groove **2724** is smaller than the width H**4** of the first adhesive receiving groove **2723**. A distance between the outer annular surface of the annular rib **2722** and the groove wall is shortened, which improves an adhesive pressure of the second adhesive receiving groove **2724**, so that the sealant within the second adhesive receiving groove **2724** further enters the gap between the outer annular surface of the annular rib **2722** and the groove wall.

[0227] Referring to FIG. **29** and FIG. **30**, in some embodiments, the housing assembly **10** is provided with the receiving cavity **1001** connected to the insert groove **1008**. The housing assembly **10** includes a limiting member **1009** disposed between the receiving cavity **1001** and the insert groove **1008**. The limiting member **1009** is configured to abut against the insert body **2721**. Specifically, the limiting member **1009** may abut against the annular rib **2722** to limit an insert depth of the insert body **2721** relative to the insert groove **1008**.

[0228] Referring to FIG. **30** to FIG. **31**, in some embodiments, when viewed from a side of the receiving cavity **1001**, at least a portion of the first adhesive receiving groove **2723** and/or the second adhesive receiving groove **2724** is not shielded by the limiting member **1009** to facilitate an application of the sealant into the first adhesive receiving groove **2723** and the second adhesive receiving groove **2724**. For example, at least a portion of the first adhesive receiving groove **2723** is not shielded by the limiting member **1009**. As another example, at least a portion of the second adhesive receiving groove **2724** is not shielded by the limiting member **1009**. For a further example, at least a portion of the first adhesive receiving groove **2723** and a portion of the second adhesive receiving groove **2724** are not shielded by the limiting member **1009**. In this way, it is ensured that the adhesive is filled from a side of the first adhesive receiving groove **2723** and/or the second adhesive receiving groove **2724** not shielded by the limiting member **1009**.

[0229] Referring to FIG. **29** and FIG. **30**, in some embodiments, when viewed from the side of the receiving cavity **1001**, an area of a region of the first adhesive receiving groove **2723** that is shielded by the limiting member **1009** is less than an area of a region that is not shielded by the limiting member **1009** and/or an area of a region of the second adhesive receiving groove **2724** that is shielded by the limiting member **1009** is less than an area of a region that is not shielded by the limiting member **1009**.

[0230] As the area of the region shielded by the limiting member **1009** is smaller than the area of the region not blocked by the limiting member **1009**, the area of the region available for the application of the adhesive is effectively increased, and the efficiency of the adhesive application is improved.

[0231] Referring to FIG. **29** and FIG. **30**, in some embodiments, a groove opening position of the region of the first adhesive receiving groove **2723** that is not shielded by the limiting member **1009** is closer to the receiving cavity **1001** compared to a groove opening position of the region of the first adhesive receiving groove **2723** that is shielded by the limiting member **1009**, and/or a groove opening position of the region of the second adhesive receiving groove **2724** that is not shielded by the limiting member **1009** is closer to the receiving cavity **1001** compared to a groove opening position of the region of the second adhesive receiving groove **2724** that is shielded by the limiting member **1009**. In this way, the adhesive application is further facilitated, and the probability of the adhesive application is increased.

[0232] In some embodiments, in a direction from a side that is never shielded by the limiting member **1009** to a side that is shielded by the limiting member **1009**, a groove bottom of the first adhesive receiving groove **2723** and/or the second adhesive receiving groove **2724** is tilted in a direction away from the receiving groove **1001**, thereby facilitating a flow of adhesive along the bottom of the groove from the side not shielded by the limiting member **1009** to the side shielded by the limiting member **1009**, thereby improving the efficiency of the adhesive application.

[0233] Referring to FIG. **29**, in some embodiments, the wearing assembly **27** further includes an elastic metal wire **273** and a wrapping member **274**. The elastic metal wire **273** and the lead assembly **271** are disposed to penetrate from an outer end surface of a side of the insert body **2721** away from the interior of the housing assembly **10** into the insert body **2721**. The wrapping member **274** is disposed to wrap the side of the insert body **2721** away from the interior of the housing assembly **10** along a circumference direction of the insert body **2721**, and to leave a side of the insert body **2721** toward the interior of the housing assembly **10** exposed.
[0234] Referring to FIG. **29** and FIG. **30**, the wrapping member **274** circumferentially wraps the insert body **2721** that is not inserted into the insert slot **1008**. The wrapping member **274** may be a rubber member or a silicone member, so as to be waterproof to protect the elastic metal wire **273**, the lead assembly **271**, and the insert body **2721** that are not inserted into the housing assembly **10**, which is conducive to improving a cosmetic quality of the headphone **100**.
[0235] The foregoing describes the movement assembly **1** and the earhook assembly **2**, and the following describes a structure of the rear hanging assembly **3** of the headphone **100**, etc.
[0236] As shown in FIG. **34**, the rear hanging assembly **3** includes the wearing assembly **30**. The wearing assembly **30** may include an elastic metal wire **31**, a lead assembly **32**, an insert member **33**, and a wrapping assembly **34**.
[0237] In some embodiments, the elastic metal wire **31** may be made of a titanium alloy, or other metals or alloys that provide an elasticity, which is not limited herein. The insert member **33** is fixed to an end of the elastic metal wire **31**.
[0238] In some embodiments, the insert member **33** further includes the insert body **331**, and the lead assembly **32** has an exposed portion **320** that is exposed from a body wrapping portion **341** and extends to the insert body **331**.
[0239] In some embodiments, at least one of the insert body **331** and the wrapping assembly **341** is configured to wrap a periphery of the exposed portion **320**.
[0240] In some embodiments, the wrapping assembly **341** is configured to wrap the periphery of the exposed portion **320**. Specifics of the wrapping assembly **341** further wrapping the periphery of the exposed portion **320** are described as follows.
[0241] In some embodiments, the wrapping assembly **34** includes the body wrapping portion **341** and an epitaxial wrapping portion **342**. The body wrapping portion **341** is configured to wrap the elastic metal wire **31** and a periphery of the lead assembly **32**.
[0242] The epitaxial wrapping portion **342** is configured to wrap the periphery of the exposed portion **320** of the lead assembly **32**, while the epitaxial wrapping portion **342** partially wraps the insert body **331** at a circumference of the insert body **331**.
[0243] The inventor of the present disclosure has found in a long-term study that, after the above-mentioned wearing assembly **30** is connected and mated with the housing assemblies **20**, **26**, etc., the adhesive may be further applied for fixing, and the adhesive often directly contacts the lead assembly **32**, and due to a poor bonding between a plastic material (e.g., Teflon) on an outer surface of the lead assembly **32** and the adhesive, on the one hand, an unstable fixing effect may be caused, and on the other hand, the sealing between a mating structure and the housing assembly **20**/**26** may also be poor. Due to a softness of the lead assembly **32**, a yield of the mating between the wearing assembly **30** and the housing assemblies **20** and **26** is low. Therefore, by adding the epitaxial wrapping portion **342** to protect the exposed portion **320**, the adhesive may not directly contact the exposed portion of the lead assembly **32**, thereby enhancing the bonding effect of the adhesive by the epitaxial covering portion **342**, the fixing effect and the sealing effect may be improved, and a mounting yield of the wearing assembly **30** may also be improved. Furthermore, the epitaxial wrapping portion **342** partially covers the insert body **331** on the circumference of the insert body **331**, which allows the insert body **331** to be partially exposed on the circumference to ensure a structural strength of the mating.
[0244] In some embodiments, as shown in FIG. **35**, a thickness D**20** of the epitaxial wrapping

portion **342** is less than a thickness D**21** of the body wrapping portion **341**, and the epitaxial wrapping portion **342** is connected to the body wrapping portion **341** and forms a step structure.

[0245] In some embodiments, the epitaxial wrapping portion **342** is integrally molded with the body wrapping portion **341**. Specifically, the epitaxial wrapping portion **342** is integrally molded with the body wrapping portion **341** by injection, molding, etc. Of course, the epitaxial wrapping portion **342** may also be disposed separately from the body wrapping portion **341**. For example, the epitaxial wrapping portion **342**, which is disposed in a tubular shape, is pre-molded, and suited to the periphery of the exposed portion **320**.

[0246] In some embodiments, a surface of the insert body **331** is disposed with a receiving groove **332**. The receiving groove **332** has an opening **333** in a cross section perpendicular to an insertion direction cz of the insert body **331**. The exposed portion **320** is disposed in the receiving groove **332**. The epitaxial wrapping portion **342** is integrally molded with the body wrapping portion **341**. When viewing from an opening direction of the opening **333** of the receiving groove **332**, at least a portion of the exposed portion **320** disposed within the receiving groove **332** has an externally visible region. The externally visible region is wrapped by the epitaxial wrapping portion **342**. In some embodiments, the length direction of the receiving groove **332** is disposed along the insertion direction cz, and it is understood that the length direction of the receiving groove **332** is the same as the insertion direction cz.

[0247] In some embodiments, the epitaxial wrapping portion **342** fills a gap between the exposed portion **320** and the groove wall of the receiving groove **332**.

[0248] In some embodiments, as shown in FIG. **35** and FIG. **40**, the wrapping assembly **34** further includes an epitaxial insert portion **343** integrally molded with the body wrapping portion **341**. The epitaxial insert portion **343** contacts an end surface of a side of the insert body **331** toward the body wrapping portion **341** toward an end surface of the side of the body wrapping portion **341**. The epitaxial insert portion **343** may be inserted into the insert groove **2001** of the housing assembly **20** with the insert body **331**. By disposing the epitaxial insert portion **343**, an insert depth of the insert body **331** may be increased, which in turn improves a structure stability.

[0249] As shown in FIG. **36**, in some embodiments, the insert member **33** further includes an epitaxial positioning portion **334** protruding at an end surface of the insert body **331**, and the epitaxial positioning portion **334** is embedded within the epitaxial insert portion **343**. By disposing the epitaxial positioning portion **334** embedded within the epitaxial insert portion **343**, an area of a contact interface between the insert body **331** and the wrapping assembly **34** is increased, thereby enhancing the stability between the insert body **331** and the wrapping assembly **34**.

[0250] In some other embodiments, when there is no epitaxial insert portion **343**, the epitaxial positioning portion **334** protrudes from the end surface of the insert body **331** toward the side of the body wrapping portion **341**, and the epitaxial positioning portion **334** is embedded in the body wrapping portion **341**, and the area of the contact interface between the inserted body **331** and the wrapping assembly **34** is likewise increased.

[0251] In some embodiments, the elastic metal wire **31** is threaded through the epitaxial positioning portion **334** to improve a fixing effect of the elastic metal wire **31** and the insert member **33**.

[0252] In some embodiments, as shown in FIG. **35**, in the cross section perpendicular to the insertion direction cz of the insert body **331**, the insert body **331** has a length direction G**2** and a width direction G**3**. The exposed portion **320** and the elastic metal wire **31** are spaced apart along the width direction G**3**. A machinability of the insert member **33** is increased by setting the exposed portion **320** and the elastic metal wire **31** spaced apart along the width direction G**3** of the insert body **331**.

[0253] As shown in FIG. **37**, in some other embodiments, the exposed portion **320** is spaced apart from the elastic metal wire **31** along the length direction G**2**. As the insert body **331** is relatively limited in size in the width direction G**3**, and by setting the exposed portion **320** spaced apart from the elastic metal wire **31** along the length direction G**2** in the insert body **331**, the space utilization

of the insert member **33** is improved.
[0254] As shown in FIG. **36**, the insert body **331** is disposed with two slots **335** spaced apart along the length direction G**2**, and the exposed portion **320** and the elastic metal wire **31** are disposed between the two slots **335** in the length direction G**2**.
[0255] In some embodiments, a covering width of the epitaxial wrapping portion **342** over the insert body **331** along a circumferential direction of the insert body **331** is less than or equal to one-fourth of a maximum circumferential length of the insert body **331**. The maximum circumferential length refers to a maximum circumference of the insert body **331** in a reference plane perpendicular to the insertion direction cz when the exposed portion **320** and the epitaxial wrapping portion are not covered.
[0256] The sealant is configured to seal a gap from at least one of the insert body **331** and the exposed portion **320** of the wrapping assembly **34** to a groove wall of the insert slot **2001**. Surface wettability of the sealant on a surface of the at least one of the insert body **331** and the exposed portion **320** of the wrapping assembly **34** is greater than surface wettability of the sealant on a surface of the lead assembly **32**.
[0257] The housing assembly **20** of the earhook assembly **2** is disposed with the insert groove **2001**, and the epitaxial wrapping portion **342** and the epitaxial insert portion **343** as described above are inserted into the insert groove **2001** with the insert body **331**. In some embodiments, the headphone **100** also includes the sealant (not shown) configured to seal the gap between the insert body **331** and the slot wall of the insert slot **2001** and the gap between the slot wall of the insert slot **2001** and the epitaxial wrapping portion **342**. The surface wettability of the sealant on the surface of the epitaxial wrapping portion **342** is greater than the surface wettability of the sealant on the surface of the lead assembly **32**.
[0258] As shown in FIG. **38**, the lead assembly **32** includes a lead body **321** and an insulating layer **322** wrapped in a periphery of the lead body **321**. In some embodiments, a material of the insulating layer **322** is polytetrafluoroethylene, and a material of the epitaxial wrapping portion **342** is silicone. The surface wettability of the sealant on the surface of the epitaxial cladding portion **342** is greater than the surface wettability of the sealant **2002** on a surface of the insulating layer **322**. In this way, a better bonding effect between the sealant and the epitaxial wrapping portion **342** is achieved, and the sealing effect and the fixing effect may be effectively improved, which in turn enables the headphone **100** to have a good waterproof effect.
[0259] One exemplary structure of the wearing assembly **30** described above mentions that the wrapping assembly **34** further wraps the exposed portion **320**. Another exemplary structure of the wearing assembly **30** where the insert body **331** further wraps the exposed portion **320** is described below.
[0260] As shown in FIG. **39**, the wearing assembly **30** may specifically include the elastic metal wire **31**, the lead assembly **32**, the insert member **33**, and a wrapping assembly **34′**. The insert member **33** is fixed to an end portion of the elastic metal wire **31**. The insert member **33** further includes the insert body **331**. The wrapping assembly **34′** includes a body wrapping portion **341′**. The body wrapping portion **341′** is configured to wrap the periphery of the elastic metal wire **31** and the lead assembly **32**. The lead assembly **32** has the exposed portion **320** that is exposed from the body wrapping portion **341′** and extends to the insert body **331**.
[0261] In some embodiments, the inset body **331** may be configured to wrap around the periphery of the exposed portion **320**. As shown in FIG. **40**, the insert body **331** is disposed with a hole path **336**. The exposed portion **320** of the lead assembly **32** penetrates the hole path **336**. For example, the hole path **336** on the insert body **331** is preformed, with the exposed portion **320** subsequently inserted into and through the hole path **336**.
[0262] By adding the hole path **336** on the insert body **331** to protect the exposed portion **320**, the adhesive may not directly contact the exposed portion of the lead assembly **32**, and the insert body **331** may contact the adhesive to enhance the bonding effect, thereby improving the fixing effect

and the sealing effect, and improving the mounting yield of the wearing assembly **30**. Furthermore, the insert body **331** may provide a better protection for the lead assembly **32** through the exposed portion **320** surrounding the hole path **336**.

[0263] In some embodiments, a hole wall of the hole path **336** closely surrounds the periphery of the exposed portion **320** along a circumferential direction of the exposed portion **320**.

[0264] In some embodiments, as shown in FIG. **39**, the wrapping assembly **34′** further includes an epitaxial wrapping portion **342′** integrally molded with the body wrapping portion **341′**. The epitaxial wrapping portion **342′** wraps an end of the insert body **331** towards the body wrapping portion **341′** along the circumferential direction of the insert body **331**. An end of the insert body **331** away from the body wrapping portion **341′** is exposed, thereby improving the waterproofing effect of an end surface of the insert body **331** toward the side of the body wrapping portion **341′**.

[0265] In the insert direction cz of the insert body **331**, a wrapping length L**6** of the epitaxial wrapping portion **342′** on the insert body **331** is in a range of 0.2 mm-2.0 mm.

[0266] Referring to the above embodiment, In some embodiments, the insert member **33** further includes the epitaxial positioning portion **334** (the epitaxial positioning portion **334** may refer to the epitaxial positioning portion illustrated in FIG. **36**). The epitaxial positioning portion **334** protrudes from an end surface of a side of the insert body **331** toward the body wrapping portion **341′**, and the epitaxial positioning portion **334** is embedded within the body wrapping portion **341′**. Similarly, the elastic metal wire **31** is threaded within the epitaxial positioning portion **334**.

[0267] Referring to the above embodiment, in the cross section perpendicular to the insertion direction cz of the insert body **331** relative to the insert groove **2001**, the insert body **331** has a length direction G**2** and a width direction G**3**, and in some embodiments, the exposed portion **320** and the elastic metal wire **31** may be spaced apart along the length direction G**2**, and the exposed portion **320** and the elastic metal wire **31** may also be spaced apart along the width direction G**3**. The machinability of the insert member **33** is increased by making the exposed portion **320** spaced apart from the elastic metal wire **31** along the width direction G**3** of the insert body **331**. Of course, an arrangement manner of the exposed portion **320** and the elastic metal wire **31** as well as a manner of disposing the slot may refer to the above embodiments. The housing assembly **20** of the earhook assembly **2** is disposed with the insert groove **2001**, and the epitaxial wrapping portion **342′** and the hole path **336** described above are inserted into the insert groove **2001** along with the insert body **331**.

[0268] In some embodiments, the headphone **100** further includes the sealant (not shown). The sealant is configured to seal a gap between at least one of the insert body **331** and the wrapping assembly **34** which wraps the exposed portion **320** and a groove wall of the insert groove **2001**. The surface wettability of the sealant on the surface of the at least one of the insert body **331** and the wrapping assembly **34** which wraps the exposed portion **320** is greater than the surface wettability of the sealant on the surface of the lead assembly **32**.

[0269] In some embodiments, the sealant is configured to seal the gap between the slot body **331** and the slot wall of the insert groove **2001**. The surface wettability of the sealant on the surface of the slot body **331** is greater than the surface wettability of the sealant on the surface of the lead assembly **32**.

[0270] As shown in FIG. **38**, the lead assembly **32** includes the lead body **321** and the insulating layer **322** wrapped around the periphery of the lead body **321**, and the surface wettability of the sealant on the surface of the epitaxial wrapping portion **342** is greater than the surface wettability of the sealant on the surface of the insulating layer **322**.

[0271] In some embodiments, the material of the insulating layer **322** may be PTFE and the material of the insert body **331** may be metal. The material of the epitaxial wrapping portion **342** may be silicone. In some embodiments, the material of the insert body **331** may be any one or a combination of one or more metals or alloys, which is not limited herein. By setting the insulating layer to be a high-temperature-resistant and insulating polytetrafluoroethylene material, the wire

body is prevented from melting the insulating layer when too much heat is generated.
[0272] The foregoing is only a portion of the embodiments of the present disclosure, which is not intended to limit the scope of protection of the present disclosure, and any equivalent device or equivalent process transformations utilizing the contents of the specification of the present disclosure and the accompanying drawings or directly or indirectly applying them in other related fields of technology are similarly included in the scope of patent protection of the present disclosure.

Claims

1. A wearing assembly comprising an elastic metal wire, a lead assembly, an insert member, and a wrapping assembly, wherein the insert member is fixed to an end of the elastic metal wire and includes an insert body, the wrapping assembly includes a body wrapping portion wrapping a periphery of the elastic metal wire and the lead assembly, the lead assembly has an exposed portion that is exposed from the body wrapping portion and extends to the insert body; at least one of-the insert body or the wrap assembly further wraps a periphery of the exposed portion.
2-9. (canceled)
10. The wearing assembly of claim 1, wherein the wrapping assembly further includes an epitaxial wrapping portion wrapping the periphery of the exposed portion of the lead assembly and partially covering the insert body in a circumferential direction of the insert body.
11. The wearing assembly of claim 10, wherein a thickness of the epitaxial wrapping portion is less than a thickness of the body wrapping portion, and the epitaxial wrapping portion is connected to the body wrapping portion to form a step structure.
12. The wearing assembly of claim 10, wherein the epitaxial wrapping portion is integrally molded with the body wrapping portion.
13. The wearing assembly of claim 10, wherein a receiving groove is disposed on a surface of the insert body, and in a cross section perpendicular to an insertion direction of the insert body relative to a first insert groove, the receiving groove has an opening, the exposed portion is disposed within the receiving groove, the epitaxial wrapping portion is integrally molded with the body wrapping portion, and at least a part of the exposed portion disposed within the receiving groove has an externally visible region when viewed from the opening of the receiving groove, and the externally visible region is wrapped by the epitaxial wrapping portion.
14. The wearing assembly of claim 10, wherein a lengthwise direction of the receiving groove is disposed along the insertion direction.
15. The wearing assembly of claim 13, wherein the epitaxial wrapping portion further fills in a gap between the exposed portion and a groove wall of the receiving groove.
16. The wearing assembly of claim 10, wherein the wrapping assembly further includes an epitaxial insert portion integrally molded with the body wrapping portion, and the epitaxial insert portion contacts an end surface of the insert body towards the body wrapping portion.
17. The wearing assembly of claim 16, wherein the insert member further includes an epitaxial positioning portion protruding from the end surface of the insert body, and the epitaxial positioning portion is embedded within the epitaxial insert portion.
18. The wearing assembly of claim 10, wherein the insert member further includes an epitaxial positioning portion protruding from an end surface of the insert body towards the body wrapping portion, and the epitaxial positioning portion is embedded within the body wrapping portion.
19. The wearing assembly of claim 18, wherein the elastic metal wire is threaded within the epitaxial positioning portion.
20. The wearing assembly of claim 10, wherein in a cross section perpendicular to an insertion direction of the insert body relative to a first insert groove, the insert body has a lengthwise direction and a widthwise direction, and the exposed portion is spaced apart from the elastic metal

wire along the lengthwise direction or the widthwise direction.

21. The wearing assembly of claim 20, wherein the insert body is disposed with two slots spaced apart in the lengthwise direction, and the exposed portion and the elastic metal wire are disposed between the two slots in the lengthwise direction.

22. The wearing assembly of claim 10, wherein a covering width of the epitaxial wrapping portion over the insert body in the circumferential direction of the insert body is less than or equal to one-fourth of a maximum circumferential length of the insert body.

23. A headphone comprising the wearing assembly of claim 1 and a housing assembly, wherein a housing assembly is disposed with a first insert groove, and the insert body is inserted in the first insert groove.

24. The headphone of claim 23, further comprising a sealant configured to seal a gap between the wrapping assembly and a groove wall of the first insert groove, wherein surface wettability of the sealant on a surface of the wrapping assembly is greater than surface wettability of the sealant on a surface of the lead assembly.

25. The headphone of claim 24, further comprising an epitaxial wrapping portion, wherein the epitaxial wrapping portion wraps the periphery of the exposed portion of the lead assembly and partially covers the insert body in a circumferential direction of the insert body, the epitaxial wrapping portion is inserted into the first insert groove with the insert body; the sealant is configured to seal gaps between the insert body as well as the epitaxial wrapping portion and the groove wall of the first insert groove, wherein the surface wettability of the sealant on a surface of the epitaxial wrapping portion is greater than surface wettability of the sealant on the surface of the lead assembly.

26. (canceled)

27. The headphone of claim 25, wherein the lead assembly includes a lead body and an insulating coating wrapping a periphery of the lead body, and the surface wettability of the sealant on the surface of the epitaxial wrapping portion is greater than surface wettability of the sealant on a surface of the insulating coating.

28. The headphone of claim 24, wherein a material of the insulating coating is polytetrafluoroethylene (PTFE), and a material of the insert body is metal.

29. The headphone of claim 27, wherein a material of the insulating coating is polytetrafluoroethylene (PTFE), and a material of the epitaxial wrapping portion is silicone.